MORITA EQUIVALENCE AND THE $\mathbb{E}_\infty$-SPIN ORIENTATION

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1. Introduction

1.1. **Rapid Summary.** The main idea of this article is to unite two points of view:

(a) A Spin-structure on a vector bundle $V \to X$ is a Morita trivialization of the Clifford algebra bundle $\mathrm{Cl}(V) \to X$.
(b) A KO-orientation is a nullhomotopy of the KO-linearized J-homomorphism or, equivalently, a trivialization of the KO-module bundle classified by the composite

$$X \xrightarrow{\xi_V} BO \xrightarrow{J} \mathrm{Pic}(\mathrm{Sp}) \xrightarrow{\otimes KO} \mathrm{Pic}(\mathrm{Mod}(KO))$$

The translation is furnished by an equivalence

$$J(V) \otimes KO \simeq K^{\mathrm{gr}} \circ \mathrm{Cl}(V)\,. \tag{1}$$

between the KO-linear J-homomorphism and the graded K-theory of the stable Clifford algebra. Thus, a Spin-structure induces a KO-orientation. We will leverage this basic translation and expand on it in a couple of ways.

1.2. **Linear Overview.** This article builds on the construction of periodic graded K-theory from [AS]. We will review the concepts as needed.

In this work, a Spin-structure shall be defined as a Morita equivalence $\mathrm{Cl}(V) \simeq \mathrm{Cl}(\mathrm{rk}\, V)$. We take this as an excuse to develop a general theory of *bimodule tangential structures*. After reviewing some basic definitions, Section 2 introduces stable tangential structures $BH(A;M) \to BO$ for pairs of a super algebra A and a finitely generated projective module M. We show that maps $X \to BH_n(A;M)$ classify a rank n vector bundle $V \to X$ together with a $\mathrm{Cl}(V-n)$-A-bimodule bundle Σ whose fibers are of *type* M.

Theorem A (Theorem 2.27)**.** *For $n \geq 1$, the classifying space for vector bundles of rank n with (A,M)-structure is $BH_n(A;M)$. The structure group is the Lie group*

$$H_n(A;M) = \left(\mathrm{Pin}_n^+ \hat{\otimes}\, \mathrm{sAut}_A(M)^{\mathrm{op}}\right)_{\mathrm{ev}}, \tag{2}$$

with its natural projection map to O_n. $BH(A,M) \to BO$ is a stable tangential structure.

In 2.4, we consider examples. Proposition 2.29 then concludes that Spin-structures are Morita equivalences $\Sigma : \mathrm{Cl}(V) \simeq \mathrm{Cl}_{\mathrm{rk}\, V}$ by setting $A = M = \mathbb{R}$.

Furthermore, we can recover all the groups of Freed-Hopkins [FH21] by looking at super division algebras.

s	0	-1	-2	-3	4	3	2	1	0^c	1^c
$D(s)$	$\mathbb{R}$	Cl_{-1}	Cl_{-2}	Cl_{-3}	$\mathbb{H}$	Cl_{+3}	Cl_{+2}	Cl_{+1}	$\mathbb{C}$	$\mathbb{C}\mathrm{l}_{-1}$
$H(s)$	Spin	Pin^+	$\mathrm{Pin}^{\tilde{c}+}$	Pin^{h-}	Spin^h	Pin^{h+}	$\mathrm{Pin}^{\tilde{c}-}$	Pin^-	Spin^c	Pin^c

A popular wrong claim is that a Spin-structure on V is equivalent to an irreducible $\mathrm{Cl}(V)$ module bundle. Using Theorem A, we can refute this claim in dimensions $\mathrm{rk}(V) \not\equiv 0 \mod 8$. This settles the question of existence of irreducible Clifford bundles completely.

Proposition 1.1. *Let V be a real vector bundle of rank n with inner product.*

(1) A graded irreducible $\mathrm{Cl}(V)$-module bundle exists if and only if V admits an $H(-n)$-structure.
(2) An ungraded irreducible $|\mathrm{Cl}(V)|$-module bundle exists if and only if V admits an $H(1-n)$-structure.

The indices are read modulo 8*. For complex Clifford modules the same statements hold with H^c and indices modulo* 2*.*

In 2.7, we provide yet another, homotopy-theoretical, proof that Spin-structures are Morita trivializations by showing that the Clifford functor can be interpreted as a categorical Postnikov truncation. In particular, the stable Clifford functor fits into a fiber sequence of $\mathbb{E}_\infty$-spaces:

$$B\,\mathrm{Spin} \to BO \xrightarrow{\mathrm{Cl}} \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan})_\infty)\,, \tag{3}$$

with target the Morita $(\infty,1)$-category of super Banach algebras, which is recalled in 2.6.

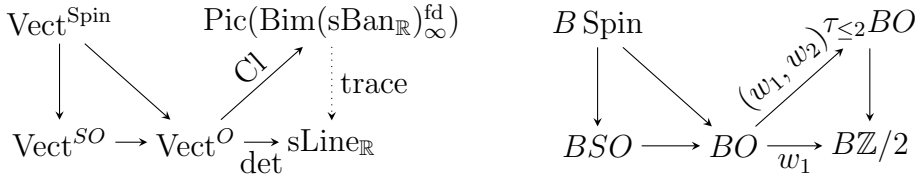


FIGURE 1. The categorical Postnikov/Whitehead tower (left) and its stable homotopical counterpart (right), with the corresponding Postnikov stages in the right-hand column of each.

Section 3 recalls briefly the construction of graded K-theory K^{gr} from [AS] and a model of the J-homomorphism to prove the following theorem.

Theorem B (Theorem 3.4)**.** *There is an invertible lax sqymmetric monoidal natural transformation making the following diagram of lax symmetric monoidal functors*

commute:

$$\begin{array}{ccc} \mathcal{I}nn & \xrightarrow{\quad J \quad} & \mathrm{Sp} \\ \downarrow{\scriptstyle \mathrm{Cl}} & & \downarrow{\scriptstyle \otimes KO} \\ \mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty & \xrightarrow{K^{gr}} & \mathrm{Mod}(KO) \end{array}$$

Here, $\mathcal{I}nn$ is the ∞-category of inner product spaces.

We will exhibit that this has powerful consequences. Using Theorem B, the fiber sequence (3) provides a nullhomotopy of the KO-linear J-homomorphism. By [ABG+14], recalled in 4.1, such nullhomotopies precisely correspond to multiplicative orientations. That is:

Corollary 1.2. *There is a map of $\mathbb{E}_\infty$-rings*

$$M\,\mathrm{Spin} \longrightarrow KO$$

We shall emphasize here that this construction is a) constructive and b) does not use functional analysis.

We furthermore prove that we can construct a lot more K-theory orientations for *all* bimodule tangential structures from Section 2 in a multiplicative manner.

Theorem C (Theorem 4.6)**.** *Let A be a super Banach algebra and let M be a finitely generated projective A-module. There are maps of spectra*

$$MH(A;M) \to K_A^{gr}\,, \tag{4}$$

satisfying the following:

(i) Let $MH(A) := \bigoplus_{[M]} MH(A;M)$, where $[M]$ runs over types of finitely generated projective A-modules. MH is the left Kan extension of the J-homomorphism along the stable Clifford functor Cl.

(ii) MH defines a lax symmetric monoidal functor $MH : \mathrm{Bim}(\mathrm{sBan})_\infty \to \mathrm{Sp}$ and there is a universal lax symmetric monoidal natural transformation $MH \Rightarrow K^{gr}$.

Therefore, the maps (4) *are multiplicative, i.e. there are coherently commutative squares*

$$\begin{array}{ccc} MH(A;M) \otimes MH(B;N) & \longrightarrow & K_A^{gr} \otimes K_B^{gr} \\ \downarrow & & \downarrow \\ MH(A\hat{\otimes}B; M\hat{\otimes}N) & \longrightarrow & K^{gr}_{A\hat{\otimes}B} \end{array}$$

In particular, inserting $A = M = \mathbb{R}$ shows that (4) *are $M\,\mathrm{Spin} \to KO$-module maps.*

(iii) If K_A^{gr} is an $\mathbb{E}_1$-KO-algebra, e.g. if A is a commutative algebra in $\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty$, then (4) defines an $H(A;M)$-orientation of the $\mathbb{E}_1$-ring spectrum K_A^{gr} if and only if $[M] \in \pi_0 K_A^{gr}$ is a unit.

Section 4.4 finally gathers examples: we can recover the tenfold way of Freed-Hopkins [FH21] and point out partial multiplicativity. We obtain multiplicative orientations $MH^G \to K^{\mathrm{gr}}_{\mathbb{R}^f[\Gamma]}$ for finite fermionic symmetry groups G and the space-time group $H^G = (\mathrm{Pin}^+_n \hat{\otimes} G)_{ev}$. For infinite discrete groups we obtain a real Bost assembly map $M\,\mathrm{Spin} \wedge B\Gamma_+ \to K_{\ell^1(\Gamma)}$. Finally, we also sketch how to obtain the Sullivan orientation $MSO \to KO[\frac{1}{2}]$ as a map of $\mathbb{E}_\infty$-rings.

1.3. **Related Work.**

2. Bimodule Bundles

2.1. **Super Algebras.** We recall the Morita theory for finite-dimensional super algebras. This material is mostly well-known or folklore, we refer to [AS] for more details. A super vector space is a $\mathbb{Z}/2$-graded $\mathbb{F}$-vector space $V = V_0 \oplus V_1$. Homomorphisms of super vector spaces are required to preserve the grading. The category of super vector spaces $\mathrm{sVect}_{\mathbb{F}}$ admits a symmetric monoidal structure $\hat{\otimes}$, whose braiding is the Koszul braiding

$$\beta : V \hat{\otimes} W \to W \hat{\otimes} V\,, \qquad v \otimes w \mapsto (-1)^{|v||w|} w \otimes v\,.$$

A super algebra is an algebra object in $(\mathrm{sVect}_{\mathbb{F}}, \hat{\otimes})$. A (graded left) module over a super algebra A is an A-module object M in $(\mathrm{sVect}, \hat{\otimes})$. Explicitly, this requires an action $A \hat{\otimes} M \to M$ that is unital and associative in the appropriate sense. We write $\mathrm{Mod}^{C_2}_A$ for the category of left modules and grading-preserving homomorphisms. An A-module M is *irreducible* if it does not contain a proper A-submodule. A super algebra is a *super division algebra* if every nonzero homogeneous element $a \in A_0 \cup A_1$ is invertible.

Lemma 2.1 (Schur's Lemma)**.** *Let M be an irreducible A-module. Then the super algebra $\mathrm{sEnd}_A(M)$ of endomorphisms of M which graded-commute with A is a super division algebra over $\mathbb{F}$.*

Remark 2.2. The super-opposite A^{op} has multiplication $a \cdot_{\mathrm{op}} b = (-1)^{|a||b|} ba$. We have $\mathrm{sEnd}_A(A) \cong A^{\mathrm{op}}$.

Lemma 2.3. *There is an equivalence between*

1. sVect-linear, $\oplus$-preserving functors $\mathrm{Mod}^{C_2}_A \to \mathrm{Mod}^{C_2}_B$.
2. B-A-bimodules M that are finitely generated projective as B-modules.

This is induced by mapping $M \mapsto M \hat{\otimes}_A -$

We write $M : A \rightsquigarrow B$, when viewing a B-A-bimodule as a morphism from A to B. There is a symmetric monoidal Morita bicategory $(\mathrm{Mor}(\mathrm{sVect}), \hat{\otimes})$ of super algebras and bimodules. A *Morita equivalence* is an isomorphism in $\mathrm{Mor}(\mathrm{sVect})$ or, equivalently, an sVect-linear additive equivalence of module categories.

Definition 2.4. Let V be a finite-dimensional real vector space with a nondegenerate symmetric bilinear form B. The *Clifford algebra* $\mathrm{Cl}(V, B)$ is defined as

the free (noncommutative) algebra on the vector space V modulo the relations $vw + wv = 2B(v,w)$ for all $v, w \in V$. We write $\mathrm{Cl}_n = \mathrm{Cl}_{+n}$ for $\mathrm{Cl}(\mathbb{R}^n, \langle , \rangle)$ and Cl_{-n} for $\mathrm{Cl}(\mathbb{R}^n, -\langle , \rangle)$, where $\langle , \rangle$ is the standard Euclidean inner product. We write $\mathbb{C}\mathrm{l}_n = \mathrm{Cl}_n \otimes \mathbb{C}$.

If A is a super algebra, we write $|A|$ for the underlying ungraded algebra.

Under the super tensor product there is a natural isomorphism of super algebras

$$\mathrm{Cl}(V,B) \hat{\otimes} \mathrm{Cl}(W,B') \cong \mathrm{Cl}(V \oplus W, B \oplus B')\,, \qquad \mathrm{Cl}(V,B)^{\mathrm{op}} \cong \mathrm{Cl}(V,-B)\,.$$

Example 2.5. We have $|\mathrm{Cl}_0| = \mathbb{R}$, $|\mathrm{Cl}_{-1}| = \mathbb{C}$, $|\mathrm{Cl}_{-2}| = \mathbb{H}$, $|\mathrm{Cl}_{+1}| \cong \mathbb{R} \oplus \mathbb{R}$.

Example 2.6. There is an isomorphism of categories $\mathrm{Mod}_A^{C_2} \cong \mathrm{Mod}_{|A \hat{\otimes} \mathrm{Cl}_{+1}|}$ [AS, ?]. For the super algebra $\mathbb{R}$ with trivial grading this amounts to the equivalence $\mathrm{sVect} \cong \mathrm{Mod}_{\mathbb{R}\oplus\mathbb{R}} \cong \mathrm{Vect}^{\times 2}$.

Example 2.7. There is a Morita equivalence $\mathbb{R} \simeq \mathrm{Cl}_{1,1}$ and furthermore $\mathrm{Cl}_{n,n} \simeq \mathbb{R}$. A bimodule inducing this equivalence is Cl_n, regarded as a left $\mathrm{Cl}_n \hat{\otimes} \mathrm{Cl}_n^{\mathrm{op}}$-module. Another nonisomorphic choice is the parity shift $\Pi\,\mathrm{Cl}_n$.

Let $(V, \langle , \rangle)$ be a inner product space. We write $\mathrm{Cl}(V\text{–}\dim V)$ for $\mathrm{Cl}(V) \hat{\otimes} \mathrm{Cl}_{-\dim V}$. By uniqueness of inner products, this super algebra is isomorphic to $\mathrm{Cl}_{\dim V, \dim V}$ and hence Morita equivalent to the ground field $\mathbb{R}$. It follows that $\mathrm{Cl}(V - \dim V)$ has two irreducible representations $E, \Pi E$ up to isomorphism.

Definition 2.8. Let A be a super algebra and M a super representation. An A-$\mathrm{Cl}(V - \dim V)$-bimodule Σ is *of type* M if the functor

$$\Sigma \hat{\otimes}_{\mathrm{Cl}(V\text{–}\dim V)} - : \mathrm{Mod}^{C_2}_{\mathrm{Cl}(V\text{–}\dim V)} \to \mathrm{Mod}^{C_2}_A\ .$$

maps the irreducible representations $\{E, \Pi E\}$ to A-modules which are isomorphic to M or ΠM. We say that M is *oriented* if M and ΠM are not isomorphic and *unoriented* if $M \cong \Pi M$.

Example 2.9. right action! Let $g \in O(\mathbb{R}^n)$. This induces an algebra homomorphism $g : \mathrm{Cl}(\mathbb{R}^n) \to \mathrm{Cl}(\mathbb{R}^n)$ and a restriction of scalars functor $g^* : \mathrm{Mod}^{C_2}_{\mathrm{Cl}_n} \to \mathrm{Mod}^{C_2}_{\mathrm{Cl}_n}$. This functor is equivalently described as tensoring with the irreducible Cl_n-bimodule ${}_g\mathrm{Cl}_n$, where the left action is restricted along g. The isomorphism class of ${}_g\mathrm{Cl}_n$ depends on $\det(g)$: We will now reason about $\mathrm{Cl}_{n,n}$-modules instead of Cl_n-bimodules. The two isomorphism classes of irreducibles are distinguished by the action of the even volume element $\omega = e_1 \dots e_n f_1 \dots f_n$ with $\omega^2 = 1$ and which satisfies $\omega a = (-1)^{|a|} a \omega$. Hence, on every representation the composition of ω with the grading operator $(-1)^F$ defines an even operator which commutes with the $\mathrm{Cl}_{n,n}$-action. If the representation is irreducible, by Schur's Lemma ω must act as $\pm(-1)^F$. The sign distinguishes the two irreducible representations. Note that the volume element of ${}_g\mathrm{Cl}_n$ is $g(e_1) \dots g(e_n) f_1 \dots f_n = \det(g)\omega$. Hence, ${}_g\mathrm{Cl}_n \cong \Pi^{\epsilon(g)}\,\mathrm{Cl}_n$ where $\epsilon(g) = (1 - \det(g))/2 \in \mathbb{Z}/2$.

Lemma 2.10. *Let Σ, Σ' be A-$\mathrm{Cl}(V - \dim V)$-bimodules. They have the same type if and only if there are bimodule isomorphisms $\Sigma \cong \Sigma'$ or $\Sigma \cong \Pi\Sigma'$.*

Proof. Suppose they have the same type. Let ${}_{\mathrm{Cl}_{n,n}}P$ and $Q_{\mathrm{Cl}_{n,n}}$ be bimodules constituting a Morita equivalence $\mathbb{R} \simeq \mathrm{Cl}_{n,n}$. P is necessarily irreducible and by assumption $\Sigma \otimes_{\mathrm{Cl}_{n,n}} P \cong M$ (or $\cong \Pi M$) which is furthermore isomorphic to $\Sigma' \otimes_{\mathrm{Cl}_{n,n}} P$. Applying $- \otimes Q$ and the isomorphism $P \otimes Q \cong \mathrm{Cl}_{n,n}$ as $\mathrm{Cl}_{n,n}$-bimodules, we obtain an isomorphism $\Sigma \cong \Sigma'$ (or $\cong \Pi\Sigma'$). The converse is trivial. □

2.2. **Fermionic Groups.**

Definition 2.11. A *fermionic group* is a group G together with a group homomorphism $|\cdot| : G \to \mathbb{Z}/2$, called *degree*, and a central element $c \neq 1$ with $c^2 = 1$ and $|c| = 0$. We can view this as a central extension:

$$\begin{array}{ccccccccc} 1 & \longrightarrow & \mathbb{Z}/2 & \longrightarrow & G & \longrightarrow & G_b & \longrightarrow & 1 \\ & & & & & \searrow^{|\cdot|} & \downarrow & & \\ & & & & & & \mathbb{Z}/2 & & \end{array}$$

G_b is called the *underlying (bosonic) group* of G. $G_{\mathrm{ev}} := \ker(|\cdot|)$ is the subgroup of even elements.

Remark 2.12. It is often beneficial to view a fermionic group as a map $BG_b \to B\mathbb{Z}/2 \times B^2\mathbb{Z}/2$, where the map to the first factor encodes $|\cdot|$ and the map to the second factor encodes the central extension. This explain the classification of fermionic group structures by group cohomology classes $w_1 \in H^1(G_b, \mathbb{Z}/2), w_2 \in H^2(G_b, \mathbb{Z}/2)$ [Ste24].

Remark 2.13. To distinguish the two occurences of $\mathbb{Z}/2$, it is customary to write $B\mathbb{Z}/2 = BO_1 = B\mathbb{Z}/2^T$ and $B^2\mathbb{Z}/2 = B^2 \operatorname{Spin}_1 = B^2\mathbb{Z}/2^F$. [Ste24]. This emphasizes the orientation (time) reversal and the internal fermionic symmetry, respectively.

Example 2.14. Let A be a super algebra. The main example of a fermionic group is $A^\times$, the set of invertible homogeneous elements in A.

Example 2.15. The group $\mathrm{Pin}_n^\pm \subset \mathrm{Cl}_{\pm n}^\times$ is generated by the unit vectors in $\mathbb{R}^n$. Its even subgroup is Spin_n, and the twisted adjoint action gives the covering maps

$$\mathrm{Pin}_n^\pm \longrightarrow O_n \quad \text{and} \quad \mathrm{Spin}_n \longrightarrow SO_n$$

with kernel $\{\pm 1\}$.

Example 2.16. Consider the canonical truncation map $BO_n \to \tau_{\leq 2} BO$. This classifies the extension $\mathbb{Z}/2 \to \mathrm{Pin}_n^+ \to O_n$ with nontrivial grading.

Proposition 2.17. *There is a fiber sequence*

$$BG_{\mathrm{ev}} \to BG_b \to \tau_{\leq 2} BO\,.$$

Proof. There are horizontal and vertical fiber sequences:

$$\begin{array}{ccccc} BK_{ev} & \longrightarrow & B(K_b)_{ev} & \longrightarrow & B^2\mathbb{Z}/2 \\ \downarrow & & \downarrow & & \downarrow \\ F & \longrightarrow & BK_b & \longrightarrow & \tau_{\leq 2}BO \\ \downarrow & & \downarrow{\scriptstyle |-|} & & \downarrow \\ * & \longrightarrow & B\mathbb{Z}/2 & = & B\mathbb{Z}/2 \end{array}$$

Hence, $BK_{\mathrm{ev}} \simeq F$. □

Definition 2.18. Let G, H be fermionic groups. The *fermionic tensor product* $G\hat{\otimes}H$ is generated by $g\hat{\otimes}h$, $g \in G$, $h \in H$ with multiplication

$$(g_1\hat{\otimes}h_1)(g_2\hat{\otimes}h_2) = c^{|g_2||h_1|}g_1g_2\hat{\otimes}h_1h_2\,,$$

subject to the relation

$$c = c_G\hat{\otimes}1 = 1\hat{\otimes}c_H\,.$$

As a set, we have $G\hat{\otimes}H = G \times H/\langle(c_G, c_H)\rangle$.

Remark 2.19. The tensor product of fermionic groups is induced from the $\mathbb{E}_\infty$-structure $\oplus$ on $B\mathbb{Z}/2 \times B^2\mathbb{Z}/2 = \tau_{\leq 2}BO$. In terms of the cohomology classes of Remark 2.12, the fermionic tensor product corresponds to $w_1(G\hat{\otimes}H) = w_1(G) + w_1(H)$ and $w_2(G\hat{\otimes}H) = w_2(G) + w_2(H) + w_1(G) \cup w_1(H)$.

Example 2.20. Consider a tensor product $A\hat{\otimes}B$ of super algebras. Then, there is a natural map $A^\times\hat{\otimes}B^\times \to (A\hat{\otimes}B)^\times$.

The *opposite fermionic group* G^{op} has multiplication $g \cdot h := (-1)^{|g||h|}hg$. In terms of classifying maps, this corresponds to the postcomposition with inversion $BG_b \to \tau_{\leq 2}BO \xrightarrow{-} \tau_{\leq 2}BO$.

Example 2.21. The fermionic group $(A^{\mathrm{op}})^\times$ is $(A^\times)^{\mathrm{op}}$.

Example 2.22. We have $(\mathrm{Pin}_n^+)^{\mathrm{op}} \cong \mathrm{Pin}_n^-$.

2.3. **Bimodule Bundles of Type** (A, M)**.** This section introduces a stable tangential structure built from bimodule bundles and identifies its classifying space.

A *locally trivial algebra bundle* is a vector bundle $B \to X$ that is locally isomorphic to a product $U \times B_0 \to U$, where B_0 is an algebra. A *locally trivial module bundle* $N \to X$ over a locally trivial algebra bundle $B \to X$ is a vector bundle with a fiberwise B-action that admits simultaneous trivializations $B|_U \cong U \times B_0$ and $N|_U \cong U \times N_0$,where N_0 is a B_0-module and the isomorphisms are compatible with the action.

Let $V \to X$ be a vector bundle equipped with an inner product $\langle,\rangle$. Applying the functor Cl to each fiber $(V_x, \langle,\rangle)$ we obtain the *Clifford algebra bundle* $\mathrm{Cl}(V) \to X$. This defines a locally trivial bundle of super algebras over X. We write $\mathrm{Cl}(V-n)$

for the super algebra bundle $\mathrm{Cl}(V - \underline{\mathbb{R}^n})$, where $\underline{\mathbb{R}^n} \to X$ is a trivial rank n vector bundle.

The fibers of the super algebra bundle $\mathrm{Cl}(V - \mathrm{rk}\, V)$ are noncanonically isomorphic to $\mathrm{Cl}_{\mathrm{rk}\, V, \mathrm{rk}\, V}$ and are hence Morita equivalent to $\mathbb{R}$. Each fiber has two inequivalent irreducible representations which are related by the parity shift operator Π.

Definition 2.23. Let A be a finite-dimensional super algebra and let M be an A-module.[1] An (A, M)-structure on a vector bundle $(V, \langle , \rangle)$ with inner product is a locally trivial $A - \mathrm{Cl}(V - \mathrm{rk}\, V)$-bimodule bundle Σ such that each fiber Σ_x is of type M. We also say that Σ is *of type* (A, M).

An isomorphism $(f, \phi) : (V, \Sigma) \to (V', \Sigma')$ of (A, M)-structures is an isometric isomorphism $f : V \to V'$ together with an isomorphism of bimodule bundles

$$\phi : \Sigma \xrightarrow{\cong} f^*\Sigma' ,$$

where the right Clifford action is pulled back along the algebra isomorphism induced by f.

We can think of the bimodule bundle as a morphism of super algebra bundles $\Sigma : \mathrm{Cl}(V - \mathrm{rk}\, V) \rightsquigarrow \underline{A}$.

Definition 2.24. We write $\mathrm{Vect}_{(A,M)}(X)$ for the groupoid of vector bundles with (A, M)-structure over X. Via pullback, this defines a functor $\mathrm{Vect}_{(A,M)} : \mathrm{Top}^{\mathrm{op}} \to \mathrm{Grpd}$.

Example 2.25 (Local model). Let $X = *$ and let V be a vector space of dimension ≥ 1. An (A, M)-structure on V is a bimodule $\Sigma : \mathrm{Cl}(V - \mathrm{rk}\, V) \rightsquigarrow A$ of type M. Any two type (A, M)-structures (V, Σ) and (V, Σ') on the same vector space are isomorphic: By Lemma 2.10, there is a bimodule isomorphism $\phi : \Sigma \cong \Sigma'$ or $\phi : \Sigma \cong \Pi\Sigma'$. In the first case, (id, ϕ) is an isomorphism. In the second case, we need to use $f \in O(V)$ with $\det(f) = -1$ since then Example 2.9 provides a bimodule isomorphism $\psi : \Pi\Sigma' \cong f^*\Sigma'$ and $(f, \psi \circ \phi) : (V, \Sigma) \to (V, \Sigma')$ is an isomorphism.

For $V = \mathbb{R}^n$, regarding Cl_n as a bimodule over itself, any (A, M)-structure is isomorphic to

$$(\mathbb{R}^n, \mathrm{Cl}_n \hat{\otimes} M) . \tag{5}$$

Any (A, M)-structure on a vector bundle of rank n is locally isomorphic to the trivial structure $(\underline{\mathbb{R}^n}, \underline{\mathrm{Cl}_n \hat{\otimes} M})$. The transition functions are continuous maps to the closed subgroup

$$\mathrm{Aut}(\mathbb{R}^n, \Sigma) = \{(f, \phi) \in O(n) \times \mathrm{GL}(\Sigma) | \phi : \Sigma \to f^*\Sigma \text{ even } A\text{-}\mathrm{Cl}_{n,n}\text{-bimodule isomorphism}\} .$$

Remark 2.26. Vector bundles with bimodule tangential structure can be added: For (V, Σ_V) in $\mathrm{Vect}_{(A,M)}(X)$ and (W, Σ_W) in $\mathrm{Vect}_{(B,N)}(X)$, there is

$$(V \oplus W, \Sigma_V \hat{\otimes} \Sigma_W) \in \mathrm{Vect}_{(A\hat{\otimes}B, M\hat{\otimes}N)}(X) . \tag{6}$$

signs

[1] By convention, M is a super vector space and a graded representation.

Theorem 2.27. *For $n \geq 1$, the classifying space for vector bundles of rank n with (A, M)-structure is $BH_n(A; M)$. For every paracompact locally compact Hausdorff space X we have*

$$\mathrm{Vect}^{\mathrm{rk}=n}_{(A,M)}(X)/\mathit{isom.} \cong \pi_0[X, BH_n(A; M)].$$

The structure group is the Lie group

$$\begin{aligned} H_n(A; M) &= \left(\mathrm{Pin}^+_n \hat{\otimes}\, \mathrm{sAut}_A(M)^{\mathrm{op}}\right)_{\mathrm{ev}} \\ &\cong \left(\mathrm{Pin}^-_n \hat{\otimes} \mathrm{sAut}_A(M)\right)_{\mathrm{ev}}., \end{aligned} \tag{7}$$

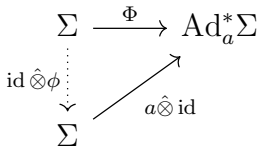
with its natural projection map to O_n. $BH(A, M) \to BO$ is a stable tangential structure, i.e. a $H_n(A; M)$-structure on V is equivalent to a $H_{n+1}(A; M)$-structure on $V \oplus \underline{\mathbb{R}}$. The tangential structure is oriented, i.e. factors as $BH(A; M) \to BSO$, if and only if M is oriented, i.e. if $M \ncong \Pi M$.

In particular, (A, M)-structures can be glued and are "homotopical". This can also be proven directly.

Proof. Let $(\mathbb{R}^n, \mathrm{Cl}_n \hat{\otimes} M)$ be the local model of a type (A, M)-structure on $\mathbb{R}^n$ (Example 2.25). There is an explicit group homomorphism to the group of grading-preserving and grading-reversing automorphisms:

$$\begin{aligned} \mathrm{Pin}^+_n \hat{\otimes} \mathrm{sAut}_A(M) &\longrightarrow \mathrm{sAut}(\mathbb{R}^n, \mathrm{Cl}(\mathbb{R}^n) \hat{\otimes} M) \\ a \hat{\otimes} \phi &\longmapsto \left[\mathrm{Ad}_a, a \hat{\otimes} \phi : \mathrm{Cl}_n \hat{\otimes} M \to \mathrm{Ad}^*_a \mathrm{Cl}_n \hat{\otimes} M\right] \end{aligned} \tag{8}$$

To show that this is an isomorphism of fermionic groups, we just note that $\mathrm{Ad_a}$ is surjective with kernel $\{\pm 1\}$ and that $\mathrm{sAut}_{A-\mathrm{Cl}_n}(M \hat{\otimes} \mathrm{Cl}_n) \cong \mathrm{sAut}_A(M)$. Write $\Sigma = M \hat{\otimes} \mathrm{Cl}(\mathbb{R}^n)$. This implies that for any $(f \in O_n, \Phi : \Sigma \to f^*\Sigma)$ there is an $a \in \mathrm{Pin}^+_n$ with $\mathrm{Ad}_a = f$ and a $\phi \in \mathrm{sAut}_A(M)$ such that $\Phi = a \hat{\otimes} \phi$:



This proves surjectivity of 8. Note that the kernel is precisely $\{1 \hat{\otimes} \mathrm{id}, -1 \hat{\otimes} -\mathrm{id}\}$, which is quotiented out in the fermionic tensor product.

We conclude that there is a homeomorphism

$$(\mathrm{Pin}^+_n \hat{\otimes} \mathrm{sAut}_A(M))_{\mathrm{ev}} \cong \mathrm{Aut}(\mathbb{R}^n, \mathrm{Cl}(\mathbb{R}^n) \hat{\otimes} M)\,,$$

as it defines continuous bijective homomorphism of Lie groups.

It follows that a rank n (A, M)-structure is patched from the trivial structure $(\mathbb{R}^n, \Sigma)$ along continuous transition functions with values in $H_n(A, M)$. That is, an (A, M)-structure is equivalent to an $H_n(A, M)$-principal bundle and $BH_n(A; M)$ is indeed a classifying space.

If $M \ncong \Pi M$ is oriented, then there is no bimodule isomorphism $\Sigma_n \to f^*\Sigma_n$ for $\det(f) = -1$, since the target is isomorphic to $\Pi\Sigma_n$. Hence the projection

$\mathrm{Aut}(\mathbb{R}^n, \Sigma_n) \to O(n)$ factors through $SO(n)$ and the tangential structure is oriented. The converse is similar.

We are left to prove stability of the tangential structure. Suppose that $V \oplus \underline{\mathbb{R}}^k$ carries an (A, M)-structure with Σ a A-$\mathrm{Cl}(V \oplus \underline{\mathbb{R}}^k - \mathrm{rk}\, V - k)$-bimodule bundle of type M. Using the Morita equivalence $\mathrm{Cl}(\underline{\mathbb{R}}^k - k) \simeq \mathbb{R}$, there also exists a A-$\mathrm{Cl}(V - \mathrm{rk}\, V)$-bimodule bundle Σ' of type M.

□

Remark 2.28. Ebert [Ebe19, Ex. 2.11] introduces *unstable* tangential bimodule structures $\theta_A(d) \to BO(d)$ via finitely generated projective $\mathrm{Cl}(V)$-A-bimodule bundles. A $\theta_A(d)$-structure on V corresponds to a bimodule tangential structure of type $(A \hat{\otimes} \mathrm{Cl}_d, N)$ with N arbitrary. Check signs!

2.4. **Equivalence and irreducible bibundles.** In this subsection we explain important instances of type (A, M)-structures. We recover the spin and pin groups and use the type conventions of Freed–Hopkins [FH21, Tables (9.24)–(9.25)].

(1) Type (A, M) for M irreducible. By Schur's lemma and Remark 2.2,

$$D = \mathrm{sEnd}_A(M)^{\mathrm{op}}$$

is a super division algebra and M is an A-D-bimodule. The functor $M \hat{\otimes}_D -$ identifies type (D, D) structures with type (A, M) structures; its inverse on these bundles is $\mathrm{sHom}_A(M, -)$. The bimodule bundle Σ is fiberwise irreducible exactly when M is irreducible.

(2) Type (A, A). Since $D_A \cong A$, Theorem 2.27 gives

$$H_n(A; A) = (\mathrm{Pin}_n^+ \hat{\otimes} A^\times)_{\mathrm{ev}}.$$

This group is denoted Spin^A by Moerss-Kamel.

(3) Type $(\mathbb{R}, \mathbb{R})$. A type $(\mathbb{R}, \mathbb{R})$ bimodule induces a fiberwise Morita equivalence $\mathrm{Cl}(V – \mathrm{rk}\, V) \simeq \mathbb{R}$. Moving the constant Clifford factor across this equivalence gives:

Proposition 2.29. *For a real vector bundle V of rank n, the following data are equivalent:*

(a) a Spin-structure on V;
(b) a type $(\mathbb{R}, \mathbb{R})$ bimodule bundle;
(c) a Morita equivalence between $\mathrm{Cl}(V)$ and the constant algebra bundle Cl_n;
(d) a fiberwise irreducible $\mathrm{Cl}(V)$-Cl_n-bimodule bundle.

Here and below the noncompact automorphism groups can be replaced by their compact reductions without changing the homotopy type of the space of structures.

(4) revise Type (D, D) for a super division algebra D. Equip each of the standard real or complex representatives with its compatible graded C^*-structure. Its compact homogeneous unit group is

$$O(D) = \{d \in D_0 \cup D_1 : d^* d = 1\}.$$

Polar decomposition gives a homotopy equivalence $D^\times \simeq O(D)$. This is the union of the unit spheres in the homogeneous summands.

For $s \in \mathbb{Z}/8$, let $D(s)$ be the real central super division algebra in the Morita class of Cl_s. For the complex types take $D^c(0) = \mathbb{C}$ and $D^c(1) = \mathbb{C}\mathrm{l}_{-1}$, following [FH21, (9.24)].

Definition 2.30. The spacetime groups are

$$H_n(s) = (\mathrm{Pin}_n^+ \hat{\otimes} O(D(s)))_{\mathrm{ev}}, \qquad H_n^c(s) = (\mathrm{Pin}_n^+ \hat{\otimes} O(D^c(s)))_{\mathrm{ev}}.$$

These are compact models for $H_n(D(s); D(s))$ and its complex analogue. They have natural maps to O_n and to $O(D(s))_b$, induced by $(K \hat{\otimes} L)_b = K_b \times L_b$.

With these conventions the ten groups are

D	H	$O(D)_b$	s
$\mathbb{R}$	Spin	$*$	0
Cl_{-1}	Pin^+	O_1	-1
Cl_{-2}	$\mathrm{Pin}^{\tilde{c}+}$	O_2	-2
Cl_{-3}	Pin^{h-}	O_3	-3
$\mathbb{H}$	Spin^h	SO_3	4
Cl_{+3}	Pin^{h+}	O_3	3
Cl_{+2}	$\mathrm{Pin}^{\tilde{c}-}$	O_2	2
Cl_{+1}	Pin^-	O_1	1
$\mathbb{C}$	Spin^c	U_1	0^c
$\mathbb{C}\mathrm{l}_{-1}$	Pin^c	$U_1 \times O_1$	1^c

(9)

We specify auxiliary virtual bundles needed below. Put $BO(D)_b = B(O(D)/\{\pm 1\})$. Let L_k be the standard bundle over BO_k, let Q be the oriented 3-plane over BSO_3, let E be the complex line over BU_1, and let λ be the line over BO_1. Define rank-zero virtual real vector bundles over $BO(D)_b$ by

$$\xi_D = \begin{cases} 0, & D = \mathbb{R}, \\ L_k \ominus k, & D = \mathrm{Cl}_{+k}, \quad k = 1,2,3, \\ k \ominus L_k, & D = \mathrm{Cl}_{-k}, \quad k = 1,2,3, \\ Q \ominus 3, & D = \mathbb{H}, \\ E_{\mathbb{R}} \ominus 2, & D = \mathbb{C}, \\ E_{\mathbb{R}} \oplus \lambda \ominus 3, & D = \mathbb{C}\mathrm{l}_{-1}. \end{cases} \tag{10}$$

Lemma 2.31 ([FH21])**.** *There is a homotopy pullback*

$$\begin{array}{ccc} BH_n(D) & \longrightarrow & B\operatorname{Spin} \\ \downarrow & & \downarrow \\ BO_n \times BO(D)_b & \xrightarrow[-\oplus\xi_D]{} & BO. \end{array}$$

Here $H_n(D) = (\operatorname{Pin}_n^+ \hat{\otimes} O(D))_{\mathrm{ev}}$*. Consequently there is a stable splitting* $BH(D) \simeq B\operatorname{Spin} \times BO(D)_b$*.*

Proposition 2.32. *For a rank* n *vector bundle* V*, the following data are equivalent, after compact reduction:*

(a) an $H_n(D)$-structure on V;
(b) a type (D, D) bimodule bundle;
(c) a fiberwise irreducible $\operatorname{Cl}(V)$-$(\operatorname{Cl}_n \hat{\otimes} D)$-bimodule bundle;
(d) a principal $O(D)_b$-bundle P and a spin structure on $V \oplus \xi_D(P)$.

Proof. The first two descriptions follow from Theorem 2.27. Dualizing a D-$\operatorname{Cl}(V-n)$-bimodule and moving the constant Cl_{-n} factor to the right gives the third description. These operations preserve irreducibility; over a super division algebra an irreducible module is the regular module up to parity. The final description is Lemma 2.31. □

Example 2.33. For $D = \operatorname{Cl}_{\mp 1}$ this is the usual description of $\operatorname{Pin}^\pm$ as a spin structure on $V \mp \det V$. Indeed, the degree condition identifies the auxiliary line with $\det V$. The obstruction is $w_2(V)$ for Pin^+ and $w_2(V) + w_1(V)^2$ for Pin^-.

Example 2.34. A Spin^c-structure consists of a complex line L and a spin structure on $V \oplus L_{\mathbb{R}}$. Its existence is equivalent to

$$w_1(V) = 0, \qquad w_2(V) = c_1(L) \bmod 2.$$

For oriented V, such a line exists exactly when $W_3(V) = \beta(w_2(V))$ vanishes.

Proof of Lemma 2.31. The fermionic group structure provides a map $\kappa_D : BO(D(s))_b \to \tau_{\leq 2} BO$. By example 2.16, the fermionic group structure on Pin_n^+ is the truncation map $BO_n \to \tau_{\leq 2} BO$. One checks that in all cases the following diagram commutes:

$$\begin{array}{ccc} BO_n \times BO(D(s))_b & \xrightarrow{(\tau, \kappa_D)} & \tau_{\leq 2} BO \times \tau_{\leq 2} BO \\ \downarrow{\scriptstyle V \oplus \xi_D} & & \downarrow{\scriptstyle \oplus} \\ BO & \longrightarrow & \tau_{\leq 2} BO \end{array} \tag{11}$$

By definition, the diagonal composite in square (11) classifies the fermionic group $\operatorname{Pin}_n^+ \hat{\otimes} O(D(s))$. Its even part is the fiber of this composite. By pullback pasting, the fiber of a composite is the pullback of the fiber of the second map. We arrive at the claim by noting that $\operatorname{fib}(BO \to \tau_{\leq 2} BO) = B\operatorname{Spin}$. □

2.5. **Existence of irreducible Clifford bundles.** The existence of an irreducible Clifford module bundle does not in general imply the existence of a spin structure. Chen [Che17] studies this question for oriented Riemannian manifolds using the negative-definite algebra $\mathrm{Cl}(-TX)$. We first give the positive-definite statement in the type convention of Definition 2.30.

Proposition 2.35. *Let V be a real vector bundle of rank n with inner product.*

(1) A graded irreducible $\mathrm{Cl}(V)$-module bundle exists if and only if V admits an $H_n(-n)$-structure.
(2) An ungraded irreducible $|\mathrm{Cl}(V)|$-module bundle exists if and only if V admits an $H_n(1-n)$-structure.

The indices are read modulo 8*. For complex Clifford modules the same statements hold with H^c_n and indices modulo* 2*.*

Proof. A graded $\mathrm{Cl}(V)$-module is a $\mathrm{Cl}(V)$-$\mathbb{R}$-bimodule. Since $\mathrm{Cl}_n \hat{\otimes} D(-n)$ is Morita equivalent to $\mathbb{R}$, the first assertion follows from Proposition 2.32.

For the second assertion, use the equivalence

$$\mathrm{Mod}_{|\mathrm{Cl}(V)|} \simeq \mathrm{Mod}^{C_2}_{\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}_{-1}}.$$

Explicitly, an ungraded module W with action ρ is sent to $W \oplus W$, with the two summands as its grading and actions

$$v \longmapsto \begin{pmatrix} 0 & \rho(v) \\ \rho(v) & 0 \end{pmatrix}, \qquad e_{-1} \longmapsto \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Conversely the invertible odd operator e_{-1} identifies the two graded summands and recovers the ungraded module. This construction is natural for orthogonal changes of frame and preserves irreducibility. Since $\mathrm{Cl}_{-1} = \mathrm{Cl}^{\mathrm{op}}_{+1}$, the resulting module is a $\mathrm{Cl}(V)$-Cl_{+1}-bimodule. The coefficient algebra $\mathrm{Cl}_n \hat{\otimes} D(s)$ has the required Morita class precisely when $n + s = 1$ in $\mathbb{Z}/8$. The complex argument is identical. □

Remark 2.36 (Negative-definite convention). The existence of a graded irreducible $\mathrm{Cl}(-V)$-module is equivalent to the existence of a $H_n(-n)$-structure, whereas the criterion for an ungraded irreducible module is a lift to $H_n(-n-1)$.

For $n = 0, 6, 7 \bmod 8$, the negative-definite ungraded types are respectively $-1, 1, 0$, corresponding to Pin^+, Pin^- and Spin. On an oriented bundle either pin structure restricts to a spin structure. This recovers [Che17, Theorem 4]. Over $\mathbb{C}$, each oriented restriction of $H^c_n(0)$ or $H^c_n(1)$ is Spin^c_n. Thus, on an oriented manifold, the existence of either a graded or an ungraded irreducible complex Clifford module bundle is equivalent to the existence of a Spin^c-structure, recovering [Che17, Theorem 2] in the ungraded case.

2.6. **The Picard groupoid of super Banach algebras.** Every finite dimensional super algebra defines a super Banach algebra by equipping it with a norm. We will now briefly recall the setup of the Morita $(\infty, 1)$-category $\mathrm{Bim}(\mathrm{sBan})_\infty$ of super Banach algebras from [AS].

Definition 2.37. Let Bim(sBan) be the symmetric monoidal $(2,1)$-category of super Banach algebras, bimodules which are finitely generated projective as a module over the target algebra and invertible intertwiners. A bimodule homotopy is $M : A \rightsquigarrow C([0,1]; B)$ regarded as a homotopy of bimodules $\mathrm{ev}_i \circ M : A \rightsquigarrow B$. A bimodule homotopy equivalence is a bimodule $M : A \rightsquigarrow B$ such that there exists a bimodule $N : B \rightsquigarrow A$ and bimodule homotopies $N \hat{\otimes}_A M \rightsquigarrow B$ and $M \hat{\otimes}_B N \rightsquigarrow A$. We let $\mathrm{Bim}(\mathrm{sBan})_\infty = \mathrm{Bim}(\mathrm{sBan})[W^{-1}]$ be the $(\infty,1)$-category obtained by localizing at bimodule homotopy equivalences.

The mapping spaces in $\mathrm{Bim}(\mathrm{sBan})_\infty$ should be interpreted as *spaces of bimodules*. By [AS] have

$$\mathrm{Hom}_{\mathrm{Bim}(\mathrm{sBan})_\infty}(A,B) \simeq \coprod_{[P]} \mathrm{Alg}(A, \mathrm{End}_B(P)^{\mathrm{op}}) /\!\!/ \mathsf{Aut}_B(P)\,, \tag{12}$$

where $[P]$ runs over isomorphism classes of finitely generated projective B-modules and where $/\!\!/$ denotes the homotopy quotient by the conjugation action on the topological space of Banach algebra homomorphisms equipped with the k-ification of the compact-open topology. If A is finite-dimensional, then this agrees with the norm topology.

2.7. **A categorical Postnikov tower.** We will now show that the Clifford algebras categorify the determinant in a precise sense. While the determinant can basically be identified with the first Stiefel-Whitney class, the Clifford algebras are a categorification of the first and second Stiefel-Whitney classes.

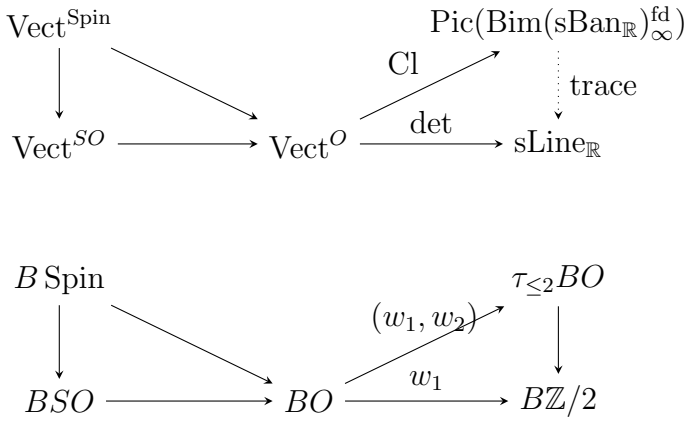


FIGURE 2. The categorical Postnikov/Whitehead tower (top) and its stable homotopical counterpart (bottom), with the corresponding Postnikov stages on the right.

The Clifford algebra construction defines a symmetric monoidal functor

$$\mathrm{Cl} : \mathrm{Vect}_{\mathbb{R}} \to \mathrm{Alg}(\mathrm{sBan}_{\mathbb{R}})_\infty \to \mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty\,. \tag{13}$$

which restricts to $\mathrm{Cl} : \mathrm{Vect}_{\mathbb{R}}^{\cong} \simeq \bigsqcup BO_n \to \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}})$ taking values in the Picard ∞-groupoid of finite-dimensional $\hat{\otimes}$-invertible algebras and invertible bimodules.

We also recall the description [AS, Thm. 3.65]: The Picard ∞-groupoid $\mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}})$ is a 2-type that has the following description: The equivalence classes of objects are generated by the Clifford algebras:

$$\pi_0 \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}}) = \{\mathrm{Cl}_0, \dots, \mathrm{Cl}_7\} \cong \mathbb{Z}/8\,.$$

The loop space $\Omega_{\mathbb{1}}(\mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}}))$ is the 1-type $\mathrm{sLine}_{\mathbb{R}}$ of real graded lines with

$$\pi_1 \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}}) = \pi_0 N^{\mathrm{hc}}(\mathrm{sLine}_{\mathbb{R}}^{\cong}) = \{\mathbb{R}, \Pi\mathbb{R}\} \cong \mathbb{Z}/2$$

$$\pi_2 \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}}) = \pi_1 N^{\mathrm{hc}}(\mathrm{sLine}_{\mathbb{R}}^{\cong}) = \pi_0 \frac{\{\text{inv. linear isomorphisms } \mathbb{R} \to \mathbb{R}\}}{\text{homotopy}} \cong \mathbb{Z}/2\,.$$

All higher homotopy groups vanish. Note that the homotopy groups to do not depend on the basepoint, since the Picard ∞-groupoid is grouplike.

We can now prove that Cl is a sort of Postnikov truncation of BO.

Proposition 2.38. *The symmetric monoidal functor* Cl *restricts to a map of* $\mathbb{E}_\infty$*-spaces*

$$\mathrm{Cl} : \mathrm{Vect}_{\mathbb{R}}^{\cong} \simeq \bigsqcup BO_n \to \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty^{\mathrm{fd}})\,.$$

On π_0 *this map is given by the mod* 8 *dimension of the vector space and induces the surjection* $\mathbb{N} \to \mathbb{Z}/8$*. It is an isomorphism on* π_1 *for all components* $n \geq 1$ *and on* π_2 *for all* $n \geq 3$*. For* $n = 2$*, the map on* π_2 *is reduction modulo* 2*,* $\mathbb{Z} \twoheadrightarrow \mathbb{Z}/2$*. The target is grouplike and hence the map extends to the group completion* $\mathrm{Vect}_{\mathbb{R}}^{\cong,\mathrm{gp}} \simeq \mathbb{Z} \times BO$*.*

Proof. The map clearly restricts. From the description of the Picard ∞-groupoid above, we are only left to prove that it induces isomorphisms on π_1, π_2. For $n \geq 1$, the two components of $\Omega BO_n = O_n$ are distinguished by the determinant. Example 2.9 identifies the Cl_n-bimodule associated to $f \in O_n$ with $\Pi^{\epsilon(f)} \mathrm{Cl}(\mathbb{R}^n)$, where $\epsilon(f) = (1 - \det(f))/2 \in \{0, 1\}$. This proves the assertion on π_1.

Put $A = \mathrm{Cl}(\mathbb{R}^n)$. Let γ_t be a path in SO_n starting at the identity, and let $\widetilde{\gamma}_t$ be the lift to $\mathrm{Spin}_n \subset \mathrm{Cl}(\mathbb{R}^n)^{\mathrm{ev}}$ with $\widetilde{\gamma}_0 = 1$. The induced algebra automorphism is $\mathrm{Cl}(\gamma_t) = \mathrm{Ad}_{\widetilde{\gamma}_t}$. Using the left-twisted bimodule convention, the associated family is

$$M_t = {}_{\mathrm{Cl}(\gamma_t)}A, \qquad a \cdot_t m \cdot b = \widetilde{\gamma}_t a \widetilde{\gamma}_t^{-1} m b.$$

Equivalently, its sections form $C([0,1]; A)$ with this pointwise action. The spin lift gives an explicit continuous trivialization by even bimodule isomorphisms

$$T_t : M_t \xrightarrow{\cong} A, \qquad T_t(m) = \widetilde{\gamma}_t^{-1} m,$$

since

$$T_t(a \cdot_t mb) = \widetilde{\gamma}_t^{-1}(\widetilde{\gamma}_t a \widetilde{\gamma}_t^{-1}) mb = a T_t(x) b.$$

The nontrivial loop in $\pi_1 SO_n$ for $n \geq 3$ lifts to a path in Spin_n with $\widetilde{\gamma}_1 = -1$, i.e. the nontrivial bimodule isomorphism $A \cong A$, which generates $\pi_2 \operatorname{Pic}(\operatorname{Bim}(\mathrm{sBan}_{\mathbb{R}})^{\mathrm{fd}}_\infty)$ (of the path component of $A = \mathrm{Cl}(\mathbb{R}^n)$). □

Corollary 2.39. *There are fiber sequence of $\mathbb{E}_\infty$-spaces*

$$B\,\mathrm{Spin} \to BO \xrightarrow{\mathrm{Cl}} \operatorname{Pic}(\operatorname{Bim}(\mathrm{sBan}_{\mathbb{R}})^{\mathrm{fd}}_\infty)\,. \tag{14}$$

$$B\,\mathrm{Spin}^c \to BO \xrightarrow{\mathbb{C}\mathrm{l}} \operatorname{Pic}(\operatorname{Bim}(\mathrm{sBan}_{\mathbb{C}})^{\mathrm{fd}}_\infty)\,. \tag{15}$$

Proof. The first fiber sequence follows directly by the definition of $B\,\mathrm{Spin}$ as a 2-connected cover of BO. The second one uses a similar description of the complex Clifford functor $\mathbb{C}\mathrm{l}$ as (w_1, W_3) into the Picard groupoid of complex super algebras. □

This explains the definition (Proposition 2.29) of a Spin-structure on V as a Morita equivalence $\mathrm{Cl}(V) \simeq \mathrm{Cl}(\mathrm{rk}\, V)$ through the lense of homotopy theory: A lift to $B\,\mathrm{Spin}$ is precisely a trivialization of $\mathrm{Cl}(V)$ in the Morita category.

3. The natural transformation $J \to K^{gr} \circ \mathrm{Cl}$

Building on the construction of graded K-theory from [AS], we will construct a monoidal natural transformation of lax symmetric monoidal functors $\mathcal{I}nn \to \mathrm{Sp}_{\geq 0}$

$$S^V \longrightarrow k^{ABS}_{\mathrm{Cl}(V)}\,, \tag{16}$$

which we compose with the map $k^{ABS} \to K^{gr} = k^{ABS}[\beta^{-1}]$ to obtain

$$J(V) = S^V \longrightarrow K^{gr}_{\mathrm{Cl}(V)}\,.$$

We then show that this induces an equivalence of KO-modules

$$S^V \otimes KO \xrightarrow{\simeq} K^{gr}_{\mathrm{Cl}(V)}\,.$$

In particular, $K^{gr} : \operatorname{Pic}(\operatorname{Bim}(\mathrm{sBan}_{\mathbb{R}})^{\mathrm{fd}}_\infty) \to \operatorname{Pic}(KO)$ is symmetric monoidal. This proves Theorem B.

Recall the definition of connective ABS K-theory k^{ABS} from [AS]: For any super Banach algebra A, we let

$$k^{ABS}_A = \mathrm{cofib}\left(k^{C_2}_{A\hat{\otimes}\,\mathrm{Cl}_{-1}} \xrightarrow{i^*_-} k^{C_2}_A\right)\,,$$

where

$$k^{C_2}_A = (\mathrm{Mod}^{C_2}_A)^{\cong,\mathrm{gp}}$$

is the group completion of the $\mathbb{E}_\infty$-monoid of finitely generated projective left A-modules and i^*_- is induced by forgetting the Cl_{-1}-action.

Check and compare with later. The Morita equivalence $\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}(-V) \simeq \mathbb{R}$ is induced by the bimodule $\mathrm{Cl}(V)$. The space of isometric embeddings $\mathcal{I}nn(\mathbb{R}^1, V)$ can be identified with the sphere $S(V) \subset V$. For any isometric embedding $f : \mathbb{R}^1 \to V$, we obtain a module by restriction along $\mathrm{Cl}_-(f) : \mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}(-\mathbb{R}) \to$

$\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}(-V)$. This leads to a bundle of modules over $S(V)$, which is constant after forgetting the Cl_{-1}-action, i.e. extends across the disk $D(V)$:

$$\underline{\mathrm{Cl}(V)} \in \mathrm{Mod}^{C_2}_{\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}_{-1}}(S(V)) \times_{\mathrm{Mod}^{C_2}_{\mathrm{Cl}(V)}(S(V))} \mathrm{Mod}^{C_2}_{\mathrm{Cl}(V)}(D(V))$$

After group completion, this is adjoint to the following square:

$$\begin{array}{ccc} \Sigma^\infty_+ S(V) & \longrightarrow & k^{C_2}_{\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}_{-1}} \\ \downarrow & & \downarrow \\ \Sigma^\infty_+ D(V) & \longrightarrow & k^{C_2}_{\mathrm{Cl}(V)} \end{array}$$

Taking cofibers, we obtain a map of spectra

$$\mathbb{S}^V \to k^{ABS}_{\mathrm{Cl}(V)}\,.$$

This describes the underlying natural transformation objectwise. It takes more care to show monoidality and naturality.

3.1. **The J-homomorphism.** Recall from [AS] that the arrow $\mathbb{R}^1 \to^{\mathrm{op}} 0$ in $\mathcal{I}nn^{\mathrm{op}}$ refines to a Smith ideal. The (co)Yoneda embedding

$$\begin{aligned} \mathcal{I}nn^{\mathrm{op}} &\longrightarrow \mathrm{Fun}(\mathcal{I}nn, \mathcal{S}) \\ V &\longmapsto [V^* : W \mapsto \mathcal{I}nn(V, W)] \end{aligned}$$

is symmetric monoidal with respect to the Day convolution symmetric monoidal structure on the target. The cofiber/quotient of a Smith ideal is a commutative algebra object. Commutative algebra objects with respect to Day convolution are lax symmetric monoidal functors.

Definition 3.1. The J-homomorphism is the (lax) symmetric monoidal functor $\mathcal{I}nn \to \mathcal{S}_*$ defined by

$$J := \mathrm{cofib}(\mathbb{R}^* \to 0^*) \in \mathsf{CAlg}(\mathrm{Fun}(\mathcal{I}nn, \mathcal{S}_*))\,. \tag{17}$$

The underlying functor of J is

$$J(V) = \mathrm{cofib}(\mathcal{I}nn(\mathbb{R}, V) \to \mathcal{I}nn(0, V)) = \mathrm{cofib}(S(V) \to *) = S^V\,.$$

$J(V)$ comes with a natural $O(V)$-action which readily identifies with the natural action on the 1-point compactification of V. Since $J(V) \wedge J(W) \simeq J(V \oplus W)$, J is strong symmetric monoidal.

Remark 3.2. We will also write J for the symmetric monoidal composite

$$\mathcal{I}nn \xrightarrow{J} \mathcal{S}_* \xrightarrow{\Sigma^\infty} \mathrm{Sp}\,.$$

3.2. **Construction.** Let $\mathcal{C}$ be a symmetric monoidal category. Then $\mathrm{Map}_{\mathcal{C}} : \mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathcal{S}$ is lax symmetric monoidal.[2] If $F : \mathcal{C} \to \mathcal{D}$ is symmetric monoidal, then $F : \mathrm{Map}_{\mathcal{C}} \to \mathrm{Map}_{\mathcal{D}} \circ (F^{\mathrm{op}} \times F)$ is a lax symmetric monoidal natural transformation.

Let $\mathcal{D}$ be a symmetric monoidal category with duals. Then there is a symmetric monoidal duality functor $(-)^\vee : \mathcal{D}^{\mathrm{op}} \to \mathcal{D}$ which assigns a dual to every object. Moreover, there is a lax symmetric monoidal natural transformation:

$$\mathrm{Map}_{\mathcal{D}}(d, d') \xrightarrow{\simeq} \mathrm{Map}_{\mathcal{D}}(\mathbb{1}, d^\vee \otimes d')\,.$$

Let $F : \mathcal{C} \to \mathcal{D}$ be symmetric monoidal and let $\mathcal{D}$ have duals. Let $G : \mathcal{D} \to \mathrm{Sp}$ be lax symmetric monoidal. We obtain a lax symmetric monoidal natural transformation

$$\mathrm{Map}_{\mathcal{C}}(c, c') \to \mathrm{Map}_{\mathcal{D}}(F(c), F(c')) \simeq \mathrm{Map}_{\mathcal{D}}(\mathbb{1}, F(c)^\vee \otimes F(c)) \to \mathrm{Map}_{\mathrm{Sp}}(G(\mathbb{1}), G(F(c)^\vee \otimes F(c')))$$

This is adjoint to a lax symmetric monoidal natural transformation of functors $\mathcal{C}^{\mathrm{op}} \times \mathcal{C} \to \mathrm{Sp}$:

$$\Sigma^\infty_+ \mathrm{Map}_{\mathcal{C}}(c, c') \to \Sigma^\infty_+ \mathrm{Map}_{\mathcal{C}}(c, c') \otimes G(\mathbb{1}) \to G(F(c)^\vee \otimes F(c')) \tag{18}$$

We consider $\mathcal{C} = \mathcal{I}nn$, $\mathcal{D} = \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})^{\mathrm{fd}}_\infty)$, $F = \mathrm{Cl}$ and $G = k^{C_2}$ to obtain a lax symmetric monoidal natural transformation of functors $\mathcal{I}nn^{\mathrm{op}} \times \mathcal{I}nn \to \mathrm{Sp}$:

$$\Sigma^\infty_+ \mathcal{I}nn(V, W) \to k^{C_2}_{\mathrm{Cl}(-V)\hat{\otimes}\,\mathrm{Cl}(W)} \tag{19}$$

Restricting along the Smith ideal $[1]_{min} \to \mathcal{I}nn^{\mathrm{op}}$ we obtain a lax symmetric monoidal natural transformation of functors $[1]_{min} \times \mathcal{I}nn \to \mathrm{Sp}$. At $V \in \mathcal{I}nn$ this is the following commutative square:

$$\begin{array}{ccc} \Sigma^\infty_+ \mathcal{I}nn(\mathbb{R}, V) & \longrightarrow & k^{C_2}_{\mathrm{Cl}(V)\hat{\otimes}\,\mathrm{Cl}_{-1}} \\ \downarrow & & \downarrow{\scriptstyle i^*_-} \\ \Sigma^\infty_+ \mathcal{I}nn(0, V) & \longrightarrow & k^{C_2}_{\mathrm{Cl}(V)} \end{array}$$

Since taking cofibers is symmetric monoidal we obtain a lax symmetric monoidal natural transformation of functors $\mathcal{I}nn \to \mathrm{Sp}$:

$$J(V) \longrightarrow k^{ABS}_{\mathrm{Cl}(V)}\,.$$

Remark 3.3. Let us unravel what the adjoint of (19) $\mathcal{I}nn(V, W) \to \Omega^\infty k^{C_2}_{\mathrm{Cl}(-V)\hat{\otimes}\,\mathrm{Cl}(W)}$ does. An isometric embedding $f : V \to W$ induces an algebra homomorphism $\mathrm{Cl}(f) : \mathrm{Cl}(V) \to \mathrm{Cl}(W)$ and a further a $\mathrm{Cl}(W)$-$\mathrm{Cl}(V)$-bimodule $\mathrm{Cl}(W)_{\mathrm{Cl}(f)}$. By duality this is equivalently a graded left $\mathrm{Cl}(-V)\hat{\otimes}\,\mathrm{Cl}(W)$-module and defines an element of $k^{C_2}_{\mathrm{Cl}(-V)\hat{\otimes}\,\mathrm{Cl}(W)}$.

[2]This follows by adjunction from the symmetric monoidal Yoneda embedding and the universal property of Day convolution $\mathrm{Fun}^{lax}(\mathcal{C}, \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})) \simeq \mathrm{Fun}^{lax}(\mathcal{C}^{\mathrm{op}} \times \mathcal{C}, \mathcal{S})$.

3.3. **Proof.**

Theorem 3.4. *There is an invertible lax symmetric monoidal invertible natural transformation making following diagram of lax symmetric monoidal functors commute:*

$$\begin{array}{ccc} \mathcal{I}nn & \xrightarrow{J} & \mathrm{Sp} \\ \downarrow{\scriptstyle \mathrm{Cl}} & & \downarrow{\scriptstyle \otimes KO} \\ \mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty & \xrightarrow{K^{gr}} & \mathrm{Mod}(KO) \end{array}$$

Upon taking cores, it restricts to a symmetric monoidal natural equivalence of symmetric monoidal functors

$$\begin{array}{ccc} \bigsqcup_n BO_n \simeq \mathcal{I}nn^{\cong} & \xrightarrow{J} & \mathrm{Pic}(\mathbb{S}) \\ \downarrow{\scriptstyle \mathrm{Cl}} & & \downarrow{\scriptstyle \otimes KO} \\ \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})^{\mathrm{fd}}_\infty) & \xrightarrow{K^{gr}} & \mathrm{Pic}(KO) \end{array}$$

Proof. We have constructed a lax symmetric monoidal natural transformation $J \to k^{ABS} \circ \mathrm{Cl}$. We can compose with the canonical map $k^{ABS} \to K^{gr}$ and KO-linearize to obtain a lax symmetric monoidal natural transformation $\alpha_V : J(V) \otimes KO \to K^{gr}(\mathrm{Cl}(V))$. Both functors are symmetric monoidal, so it suffices to show that α is an equivalence on $V = \mathbb{R}^1$. By Remark 3.3, α is induced by

$$\begin{array}{ccc} S^0 = \mathcal{I}nn(\mathbb{R},\mathbb{R}) & \longrightarrow & k^{C_2}_{\mathrm{Cl}_{+1} \hat{\otimes} \mathrm{Cl}_{-1}} \\ \downarrow & & \downarrow \\ * = \mathcal{I}nn(0,\mathbb{R}) & \longrightarrow & k^{C_2}_{\mathrm{Cl}_{+1}} \end{array}$$

and we can explicitly describe the image: The two maps $\pm\,\mathrm{id} : \mathbb{R} \to \mathbb{R}$ are mapped to the two irreducible $\mathrm{Cl}_{1,1}$-modules, cf. Example 2.9. The map $0 \to \mathbb{R}$ is mapped to Cl_{+1} as a module over itself, which is also irreducible. Using that $\mathrm{Mod}^{C_2}_{\mathrm{Cl}_{1,1}}$ is equivalent to $\mathrm{Vect} \times \mathrm{Vect}$ and that $\mathrm{Mod}^{C_2}_{\mathrm{Cl}_{+1}} \simeq \mathrm{Vect}$ via the irreducibles above, we see that the induced horizontal maps

$$\begin{array}{ccc} ko \oplus ko & \longrightarrow & k^{C_2}_{\mathrm{Cl}_{+1} \hat{\otimes} \mathrm{Cl}_{-1}} \\ \downarrow & & \downarrow \\ ko & \longrightarrow & k^{C_2}_{\mathrm{Cl}_{+1}} \end{array}$$

are equivalences. Inverting the Bott element preserves these equivalences. Upon taking vertical cofibers we obtain that α induces an equivalence $\Sigma KO \simeq K^{\mathrm{gr}}_{\mathrm{Cl}_{+1}}$. □

Remark 3.5. $\alpha_{\mathbb{R}^1}$ corresponds to an element $\alpha \in \pi_1 K^{\mathrm{gr}}(\mathrm{Cl}_1) \cong \mathbb{Z}$. This coincides with the periodicity element α constructed in [AS, Sec. 3.5].

4. Applications

4.1. **Thom spectra.** Let $\xi : X \to BO$ be the classifying map of a vector bundle and let R be a ring spectrum. We recall from [ABG$^+$14] that ξ is *R-oriented* if the composite map

$$X \to BO \xrightarrow{J} \mathrm{Pic}(\mathrm{Sp}) \xrightarrow{\otimes R} \mathrm{Pic}(R) \tag{20}$$

which classifies R-homology of the Thom space is homotopic to the constant map const_R. The datum of a nullhomotopy of (20) is equivalent to a map of spectra

$$M\xi = \operatorname*{colim}_X (J \circ \xi) \to R\,, \tag{21}$$

which is an equivalence after R-linearization. If a universal bundle $\xi : BG \to BO$ admits an R-orientation, we say conversely that R is *G-oriented*. In this case we obtain a map

$$MG := \operatorname*{colim}_{BG} J \circ \xi \to R\,. \tag{22}$$

Proposition 4.1. *Let X be an $\mathbb{E}_\infty$-space and let $\xi : X \to BO$ be an $\mathbb{E}_\infty$-map. Then $\mathbb{E}_\infty$-nullhomotopies of (20) are the same as $\mathbb{E}_\infty$-maps $M\xi \to R$.*

Remark 4.2. Let N be an R-module spectrum. Then a map $M\xi \to N$ is equivalent to a homotopy from

$$X \xrightarrow{\xi} BO \xrightarrow{J} \mathrm{Pic}(\mathrm{Sp}) \xrightarrow{\otimes R} \mathrm{Mod}(R)$$

to the constant map const_N.

We now provide an easy Corollary of Theorem 3.4.

Corollary 4.3. *Let $V \to X$ be a vector bundle and let X be a finite CW complex. Then there is an equivalence of spectra*

$$KO(Th(V)) \simeq \lim_{x \in X} K^{gr}_{\mathrm{Cl}(-V_x)} \simeq \Gamma(X; K^{gr}_{\mathrm{Cl}(-V)})$$

and there is a natural isomorphism

$$\widetilde{KO}^*(Th(V)) \cong K^{gr}_{-*}(\Gamma(X; \mathrm{Cl}(-V)))$$

with the graded K-theory of the Banach algebra $\Gamma(X; \mathrm{Cl}(-V))$.

Proof. Let $\xi : X \to BO_n$ be a classifying map for V. Then $\Sigma^\infty Th(V) = \mathrm{colim}_X J \circ \xi$ and consequently:

$$\begin{aligned} KO(Th(V)) = \mathrm{map}(\operatorname*{colim}_X J \circ \xi, KO) &\simeq \lim_X \mathrm{map}_{KO}(J \otimes KO \circ \xi, KO) \\ &\simeq \lim_X \mathrm{map}_{KO}(K^{\mathrm{gr}} \circ \mathrm{Cl} \circ \xi, KO) \\ &\simeq \lim_X K^{\mathrm{gr}} \circ \mathrm{Cl} \circ - \xi\,. \end{aligned}$$

The last part follows from [HS20, Prop. 7.1] or [BLP23, Prop. 5.30]. □

4.2. **Spin Orientation.**

Theorem 4.4. *KO (and KU) admit orientations by $\mathbb{E}_\infty$-ring homomorphisms*

$$M\,\mathrm{Spin} \to KO\,, \qquad M\,\mathrm{Spin}^c \to KU\,.$$

Moreover, a bundle is KO(KU)-oriented if and only if it admits a Spin(Spin^c)*-structure.*

Proof. By Proposition 4.1 we must supply an $\mathbb{E}_\infty$-nullhomotopy of

$$B\,\mathrm{Spin} \to BO \xrightarrow{J\otimes KO} \mathrm{Pic}(KO)\,.$$

Using the results of Corollary 2.39 and Theorem 3.4, this is supplied by commutativity of the following diagram of $\mathbb{E}_\infty$-maps:

$$\begin{array}{ccccc} B\,\mathrm{Spin} & \longrightarrow & BO & \xrightarrow{\quad J \quad} & \mathrm{Pic}(\mathrm{Sp}) \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Cl}} & & \downarrow{\scriptstyle \otimes KO} \\ * & \longrightarrow & \mathrm{Pic}(\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})) & \xrightarrow[K^{gr}]{} & \mathrm{Pic}(KO) \end{array}$$

By the dotted splitting, a KO-orientation is even equivalent to a Spin-orientation. □

Remark 4.5. Implicitly, the preceding proof constructs Thom classes for spin vector bundles. Using the Morita equivalence $\Sigma : \mathrm{Cl}(V) \rightsquigarrow \mathrm{Cl}_{\mathrm{rk}\,V}$ to a constant super algebra bundles, we obtain a $KO(X)$-linear equivalence

$$KO(Th(V)) \simeq \Gamma(X; K^{\mathrm{gr}}_{\mathrm{Cl}(-V)}) \simeq \Gamma(X; K^{\mathrm{gr}}_{\mathrm{Cl}_{-\,\mathrm{rk}\,V}}) \simeq \Sigma^{-\,\mathrm{rk}\,V} KO(X)$$

induced by a Thom class $u \in \widetilde{KO}^{\,\mathrm{rk}\,V}(Th(V))$.

4.3. **Multiplicativity and slice categories.** We will generalize the previous construction to include the tangential structures $BH(A;M)$. Note that the defining type M bimodule $\Sigma : \mathrm{Cl}(V - \dim V) \rightsquigarrow A$ defines a homotopy

$$\begin{array}{ccccc} BH(A,M) & \longrightarrow & BO & \xrightarrow{\quad J \quad} & \mathrm{Sp} \\ \downarrow & & \downarrow{\scriptstyle \mathrm{Cl}} & & \downarrow{\scriptstyle \otimes KO} \\ * & \xrightarrow[A]{} & \mathrm{Bim}(\mathrm{sBan}) & \xrightarrow[K^{gr}]{} & \mathrm{Mod}(KO) \end{array} \tag{23}$$

By virtue of Remark 4.2, this constructs the natural map (24).

Theorem 4.6. *Let A be a super Banach algebra and let M be a finitely generated projective A-module. There are maps of spectra*

$$MH(A;M) \to K^{gr}_A\,, \tag{24}$$

satisfying the following:

(i) Let $MH(A) := \bigoplus_{[M]} MH(A;M)$, where $[M]$ runs over types of finitely generated projective A-modules. MH is the left Kan extension of the J-homomorphism along the stable Clifford functor Cl.

(ii) MH defines a lax symmetric monoidal functor $MH : \mathrm{Bim}(\mathrm{sBan})_\infty \to \mathrm{Sp}$ and there is a universal lax symmetric monoidal natural transformation $MH \Rightarrow K^{gr}$.

Therefore, the maps (24) *are multiplicative, i.e. there are coherently commutative squares*

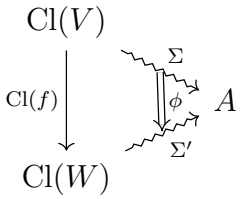
$$\begin{array}{ccc} MH(A;M) \otimes MH(B;N) & \longrightarrow & K_A^{gr} \otimes K_B^{gr} \\ \downarrow & & \downarrow \\ MH(A\hat{\otimes}B; M\hat{\otimes}N) & \longrightarrow & K^{gr}_{A\hat{\otimes}B} \end{array}$$

In particular, inserting $A = M = \mathbb{R}$ *shows that* (24) *are* $M\,\mathrm{Spin} \to KO$*-module maps.*

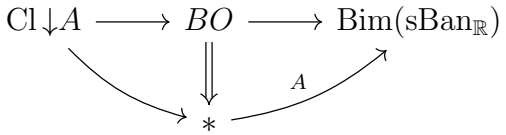
(iii) If K_A^{gr} is an $\mathbb{E}_1$-KO-algebra, e.g. if A is a commutative algebra in $\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty$, then (24) defines an $H(A;M)$-orientation of the $\mathbb{E}_1$-ring spectrum K_A^{gr} if and only if $[M] \in \pi_0 K_A^{gr}$ is a unit.

To prove assertion (i), we will exhibit a universal property of the tangential structures $BH(A;M)$ as a slice category.

Definition 4.7. Let $\mathrm{Cl}\downarrow A = BO \times_{\mathrm{Bim}(\mathrm{sBan})_\infty} (\mathrm{Bim}(\mathrm{sBan})_\infty)_{/A}$ be the slice category of the functor $\mathrm{Cl} : BO \to \mathrm{Bim}(\mathrm{sBan})_\infty$ for A a super Banach algebra. Explicitly, the objects can be thought of as pairs (V, Σ) of virtual rank zero vector spaces $V - \mathrm{rk}(V)$ together with a bimodule $\Sigma : \mathrm{Cl}(V - \mathrm{rk}\, V) \rightsquigarrow A$. A 1-morphism is a linear isometry $f : V \to W$ and an invertible bimodule map $\phi : \Sigma \to f^*\Sigma'$.



The slice category is the universal category with a natural transformation



In particular by (23), it receives maps from $BH(A, M)$ for every type $[M]$.

Proposition 4.8. *The slice category* $\mathrm{Cl}\downarrow A$ *is an* ∞*-groupoid with homotopy type*

$$\mathrm{Cl}\downarrow A \simeq \coprod_{M \text{ type}} BH(A, M)\,,$$

where M *runs over isomorphism classes of finitely generated projective* A*-modules (modulo parity shift).*

Lemma 4.9. *Let X be an ∞-groupoid and let $F : X \to \mathcal{C}$ be a functor, $c \in \mathcal{C}$. Then $F \downarrow c$ is an ∞-groupoid and*

$$F \downarrow c \simeq \operatorname*{colim}_{x \in X^{\mathrm{op}}} \mathrm{Map}_{\mathcal{C}}(F(x), c)\,.$$

Proof. The slice $\mathcal{C}_{/c} \to \mathcal{C}$ is the cartesian fibration classified by $\mathrm{Map}_{\mathcal{C}}(-, c) : \mathcal{C}^{\mathrm{op}} \to \mathrm{Spc}$. The pullback $F \downarrow c = X \times_{\mathcal{C}} \mathcal{C}_{/c}$ is the unstraightening of the composite $\mathrm{Map}_{\mathcal{C}}(F-, c)$. Unstraightening agrees with the colimit. □

To compute the slice $\mathrm{Cl}\downarrow A$ we hence have to compute a colimit over BO. Since Cl factors over $\tau_{\leq 2}BO$ we can start there.

The $\mathbb{E}_\infty$-groupoid $\mathrm{sLine}_{\mathbb{R}} \simeq \mathrm{Aut}_{\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty}(\mathbb{R})$ acts on $\mathrm{Map}_{\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty}(\mathbb{R}, A) \simeq (\mathrm{Mod}_A^{C_2})^{\cong}$ by precomposition, which is given by the tensor product

$$\begin{aligned} \mathrm{sLine}_{\mathbb{R}} \times \mathrm{Mod}_A^{C_2} &\longrightarrow \mathrm{Mod}_A^{C_2} \\ (L, M) &\longmapsto L \hat{\otimes} M\,. \end{aligned}$$

As an $\mathbb{E}_1$-groupoid, $\mathrm{sLine} \simeq \mathbb{Z}/2 \times B\{\pm 1\}$ and $B\,\mathrm{sLine} \simeq \tau_{\leq 2}BO$. The following Lemma computes the homotopy quotient of this action.

Lemma 4.10. *For every type $M \in \mathrm{Mod}_A^{C_2}$, denote the full sub-$\infty$-groupoid spanned by $\{M, \Pi M\}$ by $(\mathrm{Mod}_A^{C_2})^{\cong}_{[M]}$. Then*

$$(\mathrm{Mod}_A^{C_2})^{\cong}_{[M]} \simeq \coprod_{N \in \{M, \Pi M\}/\cong} B\,\mathrm{Aut}_A(N) \longrightarrow B\,\mathrm{sAut}_A(M)_b \tag{25}$$

is a principal $\mathrm{sLine}_{\mathbb{R}}$-"bundle" for this action. Here, the left hand side has a single component if $M \cong \Pi M$ and two components otherwise. Its classifying map $B\,\mathrm{sAut}_A(M)_b \to B\,\mathrm{sLine}_{\mathbb{R}} \simeq \tau_{\leq 2}BO$ is the one of the fermionic group $\mathrm{sAut}_A(M)$. Equivalently, the map (25) induces an equivalence

$$\operatorname*{colim}_{\tau_{\leq 2}BO} (\mathrm{Mod}_A^{C_2})^{\cong}_{[M]} \simeq B\,\mathrm{sAut}_A(M)_b\,. \tag{26}$$

The fiber sequence

$$\coprod_{N \in \{M, \Pi M\}/\cong} B\,\mathrm{Aut}_A(N) \to B\,\mathrm{sAut}_A(M)_b \to \tau_{\leq 2}BO$$

is a variant of Proposition 2.17.

Proof. We will just prove (26), as this is basically the definition of a higher principal bundle. As an $\mathbb{E}_1$-groupoid sLine is the product $\mathbb{Z}/2 \times B\{\pm 1\}$. Model $(\mathrm{Mod}_A^{C_2})^{\cong}_{[M]}$ by the topological groupoid $\mathcal{G}$ with objects $M, \Pi M$ and even A-linear isomorphisms. The topological action 2-groupoid associated to the sLine action has objects $N \in M, \Pi M$. A 1-morphisms $N \to N'$ is a pair (a, f) where $a \in \mathbb{Z}/2$ and $f : N \to \Pi^a N'$ is a grading-preserving isomorphism of A-modules. A 2-morphism is $\pm 1 : (a, f) \to (a, -f)$. The 2-groupoid is transitive, i.e. there is a single connected component. The endomorphisms (or isotropy) of the object M is the topological action groupoid $\{\pm 1\} \ltimes (\mathrm{Aut}_A(M) \sqcup \mathrm{Iso}_A(M, \Pi M))$. The action is free and the

quotient is $\mathrm{sAut}_A(M)/\{\pm 1\} = \mathrm{sAut}_A(M)_b$. So the entire topological 2-groupoid $\mathcal{G}$ is equivalent to the topological 1-groupoid $*//\,\mathrm{sAut}_A(M)_b$. We obtain

$$\operatorname*{colim}_{\tau_{\leq 2}BO}(\mathrm{Mod}_A^{C_2})^{\cong}_{[M]} = |\mathcal{G}| \simeq B\,\mathrm{sAut}_A(M)_b\,.$$

It is easy to see that these are equivalences over $\tau_{\leq 2}BO$. This yields the claim. □

Proof of Proposition 4.8. By Lemma 4.9, the slice can be expressed as a colimit over BO, i.e. as a homotopy quotient:

$$\mathrm{Cl}\!\downarrow\! A \simeq \operatorname*{colim}_{BO} \mathrm{Map}_{\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty}(\mathrm{Cl}(-), A)\,.$$

Since the stable Clifford functor Cl takes value at the tensor unit, the mapping space is just

$$\mathrm{Map}_{\mathrm{Bim}(\mathrm{sBan}_{\mathbb{R}})_\infty}(\mathbb{R}, A) \simeq (\mathrm{Mod}_A^{C_2})^{\cong} \simeq \coprod_{M\ \mathrm{fgp}} B\,\mathrm{Aut}_A(M)$$

By Proposition 2.38, the precomposition action of BO factors over $\tau_{\leq 2}BO = B\,\mathrm{sLine}_{\mathbb{R}}$. The action of a sLine on $\mathrm{Mod}_A^{C_2}$ is then given by tensor product and using Lemma 4.10, the homotopy quotient is

$$\operatorname*{colim}_{\tau_{\leq 2}BO} \coprod_{M\ \mathrm{fgp}} B\,\mathrm{Aut}_A(M) = \coprod_{M\ \mathrm{type}} B\,\mathrm{sAut}_A(M)_b\,.$$

Since colimits of spaces are universal (stable under pullback), we have

$$\mathrm{Cl}\!\downarrow\! A \simeq BO \times_{\tau_{\leq 2}BO} \coprod_{M\ \mathrm{type}} B\,\mathrm{sAut}_A(M)_b\,.$$

We are done by the definition of the fermionic tensor product $B\,\mathrm{sAut}_A(M)_b \times_{\tau_{\leq 2}BO} BO \simeq B(\mathrm{Pin}^+ \hat{\otimes}\,\mathrm{sAut}_A(M))$, which reproduces $BH(A, M)$. □

We are now ready to prove multiplicativity.

Proof of Theorem 4.6. The stable Clifford functor $\mathrm{Cl}\colon BO \to \mathrm{Bim}(\mathrm{sBan})_\infty$ is symmetric monoidal. By Lemma 4.11, applied to $\gamma = \mathrm{Cl}$ and $\mathcal{E} = \mathrm{Mod}(KO)$, the lax symmetric monoidal natural equivalence $\alpha\colon J \otimes KO \simeq K^{gr} \circ \mathrm{Cl}$ of Theorem 3.4 is adjoint to a lax symmetric monoidal transformation

$$\mathrm{Lan}_{\mathrm{Cl}}(J \otimes KO) \Longrightarrow K^{gr}\,.$$

Since $- \otimes KO$ preserves colimits, $\mathrm{Lan}_{\mathrm{Cl}}(J \otimes KO) \simeq (\mathrm{Lan}_{\mathrm{Cl}}\, J) \otimes KO$. The pointwise formula for left Kan extensions [Lur09, §4.3.3] computes $\mathrm{Lan}_{\mathrm{Cl}}\, J$ at A as a colimit over the comma ∞-category $BO \times_{\mathrm{Bim}(\mathrm{sBan})_\infty} (\mathrm{Bim}(\mathrm{sBan})_\infty)_{/A} = \mathrm{Cl}\!\downarrow\! A$. Writing $\pi_A\colon \mathrm{Cl}\!\downarrow\! A \to BO$ for the projection, Proposition 4.8 gives

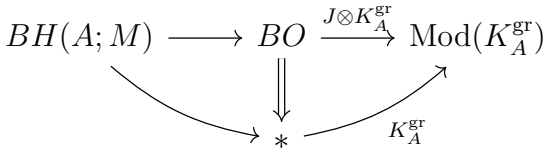
$$(\mathrm{Lan}_{\mathrm{Cl}}\, J)(A) \simeq \operatorname*{colim}_{\mathrm{Cl}\downarrow A} J \circ \pi_A \simeq \operatorname*{colim}_{\coprod_{[M]} BH(A;M)} J \simeq \bigoplus_{[M]} MH(A; M) = MH(A)\,.$$

This proves (i). The composite lax symmetric monoidal natural transformation

$$MH \Rightarrow MH \otimes KO \simeq \mathrm{Lan}_{\mathrm{Cl}}(J \otimes KO) \Rightarrow K^{gr}$$

proves (ii). It is easily seen that the map out of the component $MH(A;M)$ is induced from the cocone (23).

To prove (iii), we consider the natural transformation



obtained from (23) by K_A^{gr}-linearizing. The map $MH(A;M) \to K_A^{gr}$ is an orientation if and only if this natural transformation is an equivalence. It is an equivalence if and only if $1 \in \pi_0 K_A^{\mathrm{gr}}$ is sent to a unit. But the image of 1 is the module $M \in \pi_0 K_A^{\mathrm{gr}}$. □

Lemma 4.11. *Let $\gamma : \mathcal{C} \to \mathcal{D}$ be a symmetric monoidal functor between small symmetric monoidal categories. Let $\mathcal{E}$ be presentably symmetric monoidal. Then left Kan extension refines to an adjunction*

$$\gamma^* : \mathrm{Fun}^{\mathrm{lax}}(\mathcal{D},\mathcal{E}) \rightleftarrows \mathrm{Fun}^{\mathrm{lax}}(\mathcal{C},\mathcal{E}) : \mathrm{Lan}_\gamma$$

on the categories of lax symmetric monoidal functors and lax symmetric monoidal natural transformations.

Proof. By [Nik16, Cor. 3.8], left Kan extension is a symmetric monoidal left adjoint functor for the Day convolution symmetric monoidal structure on functor categories and γ_* is lax symmetric monoidal. By [Cno, 14.3.9] The adjunction restricts to the subcategories of commutative algebra objects, which are lax symmetric monoidal functors. □

Remark 4.12 (Size issues). Lemma 4.11 needs $\mathcal{C}$ and $\mathcal{D}$ to be small. Day convolution and the left Kan extension Lan_γ are given by colimits indexed by comma ∞-categories built from $\mathcal{C}$ and $\mathcal{D}$, and [Nik16, Cor. 3.8] is stated for small symmetric monoidal ∞-categories. In the proof of Theorem 4.6 we apply it to $\gamma = \mathrm{Cl}$, whose target $\mathrm{Bim}(\mathrm{sBan})_\infty$ is large. But $\mathrm{Bim}(\mathrm{sBan})_\infty$ is locally small by (12) and we can restrict to an arbitrary small full subcategory containing the image of Cl.

Alternatively, we can resolve this as in [Nik16, Rem. 2.8]. Fix Grothendieck universes $\mathcal{U} \in \mathcal{V}$, where *small* means $\mathcal{U}$-small. Then $\mathrm{Bim}(\mathrm{sBan})_\infty$ is $\mathcal{V}$-small. Let $\mathcal{E}$ be Sp or $\mathrm{Mod}(KO)$, and let $\widehat{\mathcal{E}}$ be the variant built from $\mathcal{V}$-small spectra. Then $\widehat{\mathcal{E}}$ is presentably symmetric monoidal relative to $\mathcal{V}$, so Lemma 4.11 applies to ev_1 with target $\widehat{\mathcal{E}}$. The inclusion $\mathcal{E} \subset \widehat{\mathcal{E}}$ is fully faithful and symmetric monoidal, and it preserves $\mathcal{U}$-small colimits. Hence lax symmetric monoidal functors to $\mathcal{E}$ form a full subcategory of those to $\widehat{\mathcal{E}}$. The functors $\tilde{J}$ and K^{gr} take values in $\mathcal{E}$. So does $\mathrm{Lan}_{\mathrm{ev}_1} \tilde{J}$, because $\mathrm{Lan}_{\mathrm{ev}_1} \tilde{J}(A)$ is a colimit over $\mathrm{Cl}{\downarrow}A$, which is essentially small. Therefore $\mathrm{Lan}_{\mathrm{ev}_1} \tilde{J} \Rightarrow K^{gr}$ is a morphism of lax symmetric monoidal functors $\mathrm{Bim}(\mathrm{sBan})_\infty \to \mathcal{E}$.

4.4. **Examples.**

4.4.1. *The tenfold way.* We now show that we can reproduce the tenfold way maps. In Section 2.4, we showed that we can recover the spacetime structure groups $H_n(s)$ of [FH21] from super division algebras as $H_n(D;D)$. By Theorem 4.6, there are M Spin-module maps

$$MH(D;D) \to K_D^{gr}$$

Using that K_D^{gr} is a shift of KO (or KU for D a complex super division algebra), we finally obtain M Spin-module maps

$$MH(s) \to \Sigma^s KO\,, \qquad MTH(s) \to \Sigma^{-s} KO\,,$$

as well as

$$MH^c(s) \to \Sigma^s KU\,, \qquad MTH(s) \to \Sigma^{-s} KU\,.$$

We use the group labels and Clifford types of [FH21, Tables (9.24)–(9.25)]. The last columns below give the factors $X(s)$ and $X^c(s)$ of the *tangential* Thom spectra $MTH(s)$ and $MTH^c(s)$. Here MO_k is the Thom spectrum of the actual universal k-plane bundle, MTO_k that of its negative, and MU_1 is the Thom spectrum of the universal complex line, of real rank 2.

s	H	K	Cartan	D	$O(D)_b$	$X(s)$
0	Spin	O_1	D	$\mathbb{R}$	$*$	$\mathbb{S}$
-1	Pin^+	O_1	DIII	Cl_{-1}	O_1	ΣMTO_1
-2	$\mathrm{Pin}^{\tilde{c}+}$	U_1	AII	Cl_{-2}	O_2	$\Sigma^2 MTO_2$
-3	Pin^{h-}	Sp_1	CII	Cl_{-3}	O_3	$\Sigma^3 MTO_3$
4	Spin^h	Sp_1	C	$\mathbb{H}$	SO_3	$\Sigma^{-3} MSO_3$
3	Pin^{h+}	Sp_1	CI	Cl_{+3}	O_3	$\Sigma^{-3} MO_3$
2	$\mathrm{Pin}^{\tilde{c}-}$	U_1	AI	Cl_{+2}	O_2	$\Sigma^{-2} MO_2$
1	Pin^-	O_1	BDI	Cl_{+1}	O_1	$\Sigma^{-1} MO_1$

(27)

s	H^c	K	Cartan	D^c	$O(D^c)_b$	$X^c(s)$
0	Spin^c	U_1	A	$\mathbb{C}$	U_1	$\Sigma^{-2} MU_1$
1	Pin^c	U_1	AIII	$\mathbb{C}\mathrm{l}_{-1}$	$U_1 \times O_1$	$\Sigma^{-3}(MU_1 \wedge MO_1)$

(28)

For a stable structure map $X \to BO$, write

$$MX = \operatorname*{colim}_X J, \qquad MTX = \operatorname*{colim}_X (-J).$$

Thus MX uses the universal rank-zero bundle, whereas MTX uses its negative. Under the splitting of Lemma 2.31, that bundle on $BH(D)$ is $W - \xi_D$. Consequently

$$MH(D) \simeq M\,\mathrm{Spin} \wedge B_D^{-\xi_D}, \qquad MTH(D) \simeq M\,\mathrm{Spin} \wedge B_D^{\xi_D},$$

where B_D^η denotes the Thom spectrum of a virtual bundle η over B_D. In the second equivalence we also use that negation of the universal stable spin bundle identifies $MT\,\mathrm{Spin}$ with $M\,\mathrm{Spin}$.

Lemma 4.13 ([FH21, (10.2)–(10.3)])**.** *The tangential Thom spectra split as M Spin-modules:*

$$MTH(s) \simeq M\,\mathrm{Spin} \wedge X(s), \qquad MTH^c(s) \simeq M\,\mathrm{Spin} \wedge X^c(s),$$

with factors as in (27) *and* (28)*. We have $MH(s) \simeq MTH(-s)$.*

Proof. Under the stable splitting of Lemma 2.31 $BH(s) \simeq B\,\mathrm{Spin} \times BO(D(s))_b$ compatible with J. Therefore

$$MH(s) = \operatorname*{colim}_{BH(s)} J \simeq \operatorname*{colim}_{B\,\mathrm{Spin}} J \wedge \operatorname*{colim}_{BO(D(s))_b} J \simeq M\,\mathrm{Spin} \wedge X(s)\,.$$

□

The M Spin-module structure gives a factorization

$$M\,\mathrm{Spin} \wedge X(s) \longrightarrow ko \wedge X(s) \longrightarrow \Sigma^{-s} KO.$$

Each $X(s)$ is the Thom spectrum of a rank-zero virtual bundle over a CW complex and is connective. Since $M\,\mathrm{Spin} \to ko$ induces isomorphisms on homotopy groups in degrees below 8, the first map does so as well.

Remark 4.14. We emphasize here a point made by Stehouwer: K-theory measures free fermion phases and is 8-periodic in the spacetime dimension. The (Anderson dual of) the spectrum $MH(s)$ corresponds to interacting phases in spacetime dimension s. Freed and Hopkins make $H(s)$ artificially periodic in s, which is not necessarily physically expected.

We now outline a multiplicativity deficit of the periodicity in s, namely only a partial multiplicativity. The multiplicative structure is

$$MH(\mathrm{Cl}_s; S) \otimes MH(\mathrm{Cl}_t; T) \to MH(\mathrm{Cl}_s \hat{\otimes} \mathrm{Cl}_t; S\hat{\otimes}T)\,.$$

Suppose S, T are irreducible. For normalized indices $s, t \in \{-3, \dots, 4\}$, a dimension argument shows that $S\hat{\otimes}T$ is irreducible precisely in the following cases, up to exchanging s and t:

$$\begin{aligned} &(s,t) = (0,u) \quad \text{for } u \in \{-3,\dots,4\}, \\ \text{or} \quad &(s,t) \in \{(-2,-1),(-1,-1),(-1,4),(1,1),(1,2),(1,4)\}. \end{aligned}$$

This gives a partial multiplicativity, e.g. for $(s,t) = (1,4)$

$$M\,\mathrm{Pin}^+ \otimes M\,\mathrm{Spin}^h \to M\,\mathrm{Pin}^{h+}\,.$$

If S, T are the left regular module, this is always multiplicative, but then $MH(\mathrm{Cl}_s; S) \simeq MH(s)$ only for $-3 \le s \le 3$.

Remark: multiplicativity seems to be related to Kaluza-Klein compactifications. Bott spiral uses finite fermionic groups.

Remark 4.15. The (unstable) multiplicativity $MTH_n(A; M) \otimes MTH_m(B; N) \to MTH_{n+m}(A\hat{\otimes}B; M\hat{\otimes}N)$ should physically be related to Kaluza-Klein compactification of TQFTs as discussed in [**?**, §6 & §7.3].

Remark 4.16 (Dimension shift in Ebert's convention). Check what this should say. Let V_d be the universal rank d bundle over $BH_d(A;M)$. Ebert's $MT\theta_B(d)$ is the Thom spectrum of $-V_d$, of virtual rank $-d$; our stable tangential convention uses $-(V_d - d) = d - V_d$. Hence, on this component,

$$BH_d(A;M)^{d-V_d} \simeq \Sigma^d BH_d(A;M)^{-V_d}.$$

Pulling back Ebert's topological index [Ebe19, Definition 2.22] and suspending by d gives

$$BH_d(A;M)^{d-V_d} \longrightarrow \Sigma^d K(A^{\mathrm{op}} \hat{\otimes} \mathrm{Cl}_{-d}) \simeq K(A^{\mathrm{op}}).$$

Here K denotes the K-theory spectrum in Ebert's convention. For $A = M = D(s)$ the coefficient algebra $A^{\mathrm{op}} \hat{\otimes} \mathrm{Cl}_{-d}$ has Clifford Morita type $-s-d$, so the target before suspension is $\Sigma^{-s-d}KO$ and after suspension it is $\Sigma^{-s}KO$. The same calculation over $\mathbb{C}$ gives $\Sigma^{-s}KU$. Thus the degree agrees with the tangential orientation above; the super-opposite accounts for $-s$ and the rank normalization accounts for d. This comparison identifies the structures and degrees; identifying the orientation maps also requires matching their Thom classes under the chosen Morita equivalences.

The $M\,\mathrm{Spin}$-module structure gives a factorization

$$M\,\mathrm{Spin} \wedge X(s) \longrightarrow ko \wedge X(s) \longrightarrow \Sigma^{-s} KO.$$

Each $X(s)$ is the Thom spectrum of a rank-zero virtual bundle over a CW complex and is connective. Since $M\,\mathrm{Spin} \to ko$ induces isomorphisms on homotopy groups in degrees below 8, the first map does so as well.

4.4.2. *Group algebras and assembly.* Let G be a finite fermionic group with central element c. The fermionic group algebra is the finite dimensional semisimple super algebra $\mathbb{R}^f[G] = \mathbb{R}[G]/(c = -1)$. $G \mapsto \mathbb{R}^f[G]$ is the symmetric monoidal left adjoint to $A \mapsto A^\times$. The adjunction unit is the inclusion $G \to \mathbb{R}^f[G]^\times$. Let $H_n^G = (\mathrm{Pin}_n^+ \hat{\otimes} G)_{ev}$. The adjunction unit induces $H_n^G \to H(\mathbb{R}[G], \mathbb{R}[G])$ and we obtain maps

$$MH^G \to MH(\mathbb{R}^f[G]; \mathbb{R}^f[G]) \to K^{gr}_{\mathbb{R}^f[G]}$$

These maps are multiplicative in G. In particular, if G is abelian, then we obtain an $\mathbb{E}_\infty$-orientation.

For purely even infinite discrete groups Γ, we can instead consider $\ell^1(G)$ and the trivial extension to a fermionic group $\hat{\Gamma} = \Gamma \times \mathbb{Z}/2$. In this case, there is a splitting $H_n^{\hat{\Gamma}} = \mathrm{Spin}_n \times \Gamma$ and hence a map

$$M\,\mathrm{Spin} \wedge B\Gamma_+ \to KO \wedge B\Gamma_+ \to K_{\ell^1(G)}\,.$$

The second map should be the real Bost assembly map and an index theorem should conjecturally link it to the index of the Dirac operator twisted by the Miscenko-Fomenko bundle. (remark 7.3LNS,)

4.4.3. *Sullivan orientation.* Consider the inclusions

$$M_{2^n}(\mathbb{R}) \to M_{2^{n+1}}(\mathbb{R})\,, \qquad T \mapsto T \oplus T\,,$$

where the ungraded Banach algebras carry the unique C^*-norm and let $A = \operatorname{colim}_{\mathbb{N}} M_{2^n}(\mathbb{R})$, the Banach completion of the union in the supremum norm. This is a special case of a UHF algebra, the CAR algebra. By [AS], K-theory preserves filtered colimits, and the transition maps act as multiplication by two, so

$$K_A = \operatorname*{colim}_{\mathbb{N}} \left(KO \xrightarrow{\cdot 2} KO \to \dots \right) = KO[\tfrac{1}{2}]\,.$$

Note that we may write $M_{2^n}(\mathbb{R}) = M_2(\mathbb{R})^{\otimes n}$ and $A = \bigotimes_{i=1}^{\infty} M_2(\mathbb{R})$ with the transition maps becoming $M_2(\mathbb{R}) \otimes \mathbb{R} \to M_2(\mathbb{R}) \otimes M_2(\mathbb{R})$. The unique algebra homomorphism $\eta : \mathbb{R} \to A$ is an idempotent in $\mathrm{Bim}(\mathrm{sBan})_\infty$, i.e. $\eta \otimes 1 : A \cong A \otimes_\pi A$. By [Lur, Prop. 4.8.2.9], A carries an essentially unique $\mathbb{E}_\infty$-structure with unit η. By Theorem 4.6, there is an $\mathbb{E}_\infty$-orientation

$$MH(A;A) \to KO[\tfrac{1}{2}]\,. \tag{29}$$

We have $H_r = H_r(A;A) = (\mathrm{Pin}_r^+ \hat{\otimes} A^\times)_{\mathrm{ev}} = \mathrm{Spin}_r \times A^\times / \langle (-1,-1) \rangle$ fitting into a short exact sequence

$$1 \to A^\times \to H_r \to SO(r) \to 1\,.$$

There is an explicit splitting using Clifford algebras: Let $\tilde{g} \in \mathrm{Spin}(r)$ be a lift of $g \in SO(r)$. The standard left action ρ_r of $\mathrm{Spin}(r)$ on $|\,\mathrm{Cl}_r\,| \cong \mathbb{R}^{2^r}$ supplies a splitting

$$\begin{aligned} SO(r) &\to H_r \\ g &\mapsto [\tilde{g}, \rho_r(\tilde{g})]\,. \end{aligned}$$

We conjecture that this lifts to an $\mathbb{E}_\infty$-splitting. In particular, there is also a (non-canonical) splitting $BH(A;A) \simeq BSO \times BA^\times$. This can also be obtained by obstruction theory. Furthermore, splittings of the fiber sequence $BA^\times \to BH(A;A) \to BSO$ are in bijection with orientations

$$MSO \to KO[\tfrac{1}{2}]\,, \tag{30}$$

which map $1 \in \pi_0 MSO \cong \mathbb{Z}$ to $1 \in \pi_0 KO[\frac{1}{2}] \cong \mathbb{Z}[\frac{1}{2}]$. For multiplicative orientations this condition is automatic. We state the $\mathbb{E}_\infty$-version:

Proposition 4.17. *The space of $\mathbb{E}_\infty$-sections of $p : BH(A;A) \to BSO$ is equivalent to the space of $\mathbb{E}_\infty$-orientations $MSO \to KO[\frac{1}{2}]$. Equivalently, $BH(A;A) \simeq \mathrm{fib}(BSO \to BSL_1(KO[\frac{1}{2}]))$.*

Proof sketch. Write $j : BSO \to BSL_1(KO[\frac{1}{2}]) = \tau_{\geq 2} BGL_1(KO[\frac{1}{2}])$ for the J-homomorphism. The space of $\mathbb{E}_\infty$-orientations is the space of $\mathbb{E}_\infty$-sections of $\mathrm{fib}(j) \to BSO$. The orientation (29) provides a map $BH(A;A) \to \mathrm{fib}(j)$ over

BSO. To show that this is an equivalence we consider the induced map on fiber sequences:

$$\begin{array}{ccccc} BA^\times & \longrightarrow & BH(A;A) & \longrightarrow & BSO \\ \downarrow & & \downarrow & & \| \\ SL_1(KO[\frac{1}{2}]) & \longrightarrow & \mathrm{fib}(j) & \longrightarrow & BSO \end{array}$$

We show that the map is an equivalence on fibers and hence on total spaces. Now $\Omega^\infty KO[\frac{1}{2}] = \Omega^\infty K_A = K_0(A) \times B\,\mathrm{GL}_\infty(A)$. We also have that $A^\times \to \mathrm{GL}_\infty(A)$ is an equivalence. Hence the underlying space of $\mathrm{GL}_1(KO[\frac{1}{2}])$ is $\mathbb{Z}[\frac{1}{2}]^\times \times BA^\times$ and $BA^\times \to \mathrm{SL}_1(KO[\frac{1}{2}])$ is an equivalence. □

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